

Banking Networks, Systemic Risk, and the Credit Cycle in Emerging Markets

Sanjiv R. Das
Santa Clara University

Madhu Kalimipalli
Wilfrid Laurier University

Subhankar Nayak
Wilfrid Laurier University *

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Abstract

We undertake a large-scale empirical examination of systemic risk among 1048 financial institutions in a large sample of 23 emerging markets, broken down into 5 regions. This work extends the large literature on systemic risk in the US, Europe, and other developed countries to emerging markets, which is relatively under-researched. We present a novel systemic risk score for each financial system by region, across time. This measure is additively decomposable and attributable to each financial institution, and may be used as an objective and quantifiable measure of whether a bank is a SIFI (systemically important financial institution). The level and timing of systemic risk is heterogeneous across the 5 regions, and this risk is concentrated in a few banks, more so pre-crisis than post-crisis. Credit and network effects account for over 2/3 of systemic risk (and in some cases, almost all of the risk), and the remaining comes from individual bank variables. Spillovers of systemic risks across regions are mostly contemporaneous within the quarter. A primary principal component (default level) accounts for 1/2 of the variation in systemic risk across the regions with the next two principal components (policy uncertainty and liquidity) accounting for 1/5 each. Aggregate default risk in a region is statistically predictable using our systemic risk metric, thereby supporting timely macro-prudential policy-making.

JEL classification: G00, G01, G10, G15

Keywords: systemic risk, network risks, default risk, emerging markets, financial institutions

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1 Introduction

Using a model of systemic risk, we provide new evidence on the interaction of credit cycles and systemic risk in emerging markets. There is limited prior literature on systemic risk and default in emerging markets. We construct bank networks to quantify the interconnectedness of major financial institutions in five regions of emerging markets, and therefrom compute a systemic risk measure and network metrics. We show that these metrics are useful in predicting overall default levels in these emerging markets in future quarters.¹

Starting in 2007, emerging economies accumulated significant external debt as non-financial corporations in these markets increased their external borrowing significantly through the offshore issuance of debt securities.² For example, emerging market corporate loans and debt rose from 73% of GDP at the end of 2007 to 107% of GDP by the end of 2014.³ Although greater leverage can facilitate higher corporate investment and perhaps stimulate growth, the continued accumulation of corporate debt can be concerning because many financial crises in emerging markets have been preceded by rapid leverage growth.⁴ Emerging market credit in general is dominated by bank loans. Excessive corporate leverage can lead to increased risk exposure for local banks. If the high leverage through foreign debt is not adequately hedged by emerging market firms, it can further exacerbate the risks to domestic banks. Such increased risk exposure of banks can be critical in the face of commodity and currency market shocks and global monetary policy developments (e.g., the U.S. QE taper-tantrum).

Systemic risk is defined as the risk of substantial damage to, or failure of, the financial system in a country. This is different from systematic risk, characterized by correlation amongst assets in an economy induced by a set of common factors. Whereas systematic risk is driven by unconditional correlation, systemic risk is an artifact of conditional correlation, specifically the conditional failure of the system at large driven by (or conditional on) the failure of key financial institutions in an economy, consistent with the definition in [Li and Zinna \(2014\)](#). Network effects underlie systemic risk and impact the amplitude of the credit cycle, which we define as the levels of ex ante aggregate default risk in the financial sector, measured by credit spreads. In this paper, we model systemic risk by modeling a network among banks in a country. The network provides the mechanism for transmission of risk, and is the driving force of contagion. The interconnectedness of banks described by a network is augmented with information on the credit quality of banks. We combine network and credit information into a single measure of systemic risk for the entire financial system. This measure is a modification of the model from [Das \(2016\)](#). We calculate the measure for each quarter from 2004 Q3 to 2016 Q4, a total of 50 quarters. This time series is then used for further analyses.

We calculate several metrics as part of this analysis. We compute various measures of

¹This result complements [Pagano and Sedunov \(2016\)](#), who find that a CoVaR-based measure of systemic risk is predictive of sovereign default levels in the 15 largest European countries.

²Committee on International Policy Reform: “Corporate Debt in Emerging Economies: A Threat to Financial Stability?” September, 2015; [Avdjiev et al. \(2014\)](#).

³Including the credit extended by “shadow banks”, there was even steeper rise and a higher total burden amounting to 127% of GDP (source: Economist, Nov 14, 2015). Overall the corporate debt of non-financial firms across major emerging market economies quadrupled between 2004 and 2014 (Corporate leverage in emerging markets – a concern? Global Financial Stability Report: Vulnerabilities, Legacies, and Policy, IMF, October, 2015).

⁴Increases in corporate leverage in the post-crisis period significantly increased the default risks of emerging market firms ([Dodd et al., 2019](#)).

the mathematical properties of the network each quarter such as the *diameter* of the network, because contagion travels further when diameter is low; average *degree* of the network, degree being the number of connections of each node, higher the degree greater the network interconnectedness; *fragility* or how susceptible the network is to a local problem becoming a global one; *degree HHI*, where the Herfindahl index of node degree describes the extent of concentration in the network (more concentrated networks support contagion because of their hub and spoke shape). We also report the number of *clusters*, and the cluster HHI, where a cluster is an independent group of nodes that is not connected to any other group of nodes. The greater the number of disconnected clusters, the less likely we might have economic contagion, but the more concentrated nodes are in a single cluster we have a greater chance of contagion and systemic risk.

For each quarter we also compute risk measures by bank. We have the 12 month *probability of default* (PD) of each bank; banks with high PD and high interconnectedness pose a threat to the system, so we retain the *degree* of each bank. We also calculate *betweenness centrality* for each bank in the network, which is a measure of how central a position the bank has (this is defined formally later in the modeling section). Finally, we calculate the total systemic risk for each quarter, and decompose it into the risk contributed by each bank, which offers us a metric for how systemically important a bank is. This systemic risk decomposition may be used to identify SIFIs (designated systemically important financial institutions, stipulated by the Dodd-Frank Act, 2010).

Our sample of emerging countries is obtained by combining the IMF’s and MSCI’s lists of emerging countries with firm-level data from two different sources: (1) stock return, financial and balance sheet data are extracted from Datastream, and (2) Probability of default data sourced from the Credit Risk Initiative (CRI), National University of Singapore (NUS). We employ active financial firms trading in a primary exchange in the local market, by matching the industry classification based on the Compustat Global Database. We exclude financial subsidiaries of non-financial corporations and specialized investment vehicles such as funds, REITs, and securitized assets. Our final quarterly data sample consists of 1048 financial firms comprised of 539 Banks, 389 Broker-Dealers, and 120 Insurers from 23 emerging market countries clustered into 5 regions for the period 2004-2016.

Our main results are as follows:

1. We observe considerable heterogeneity in evolution of systemic risks across the emerging markets.
 - Eastern Europe and South America have the highest systemic risks compared to other regions over time mainly during the 2004-2006 pre-crisis and 2007-2009 crisis periods. East Asia registers highest systemic risk during 2013 Taper-tantrum period and again during the 2015-2016 foreign exchange crisis period.
 - There is a high degree of concentration in systemic risks among major banks in each region, consistent with the large-scale results for developed countries in [Laeven et al. \(2016\)](#). The top 10 percentile contributors contribute 16% to 47% of the systemic variation. Eastern Europe and South America have the maximum concentration (over 46%) of systemic risk among the top 10 contributors. Most of the systemic risk for each geographic region is concentrated among fewer banks in the pre-crisis periods compared to post-crisis period.
 - Using time series and panel data regressions, we examine to what extent emerging market systemic risks across geographic regions can be explained by different risks.

Credit and network risks together explain the majority of the variation in systemic risks i.e. between 81%-94% of the time-series variation, and 64%-82% across firms and over time. Firm-specific attributes (such as leverage, profitability, loans to assets, loans to deposits, and market to book ratios) add an additional 5%-15% explanatory power in panel data regressions.

2. Examining the raw correlations among systemic risks across different geographic regions, we find that both contemporaneous and lagged effects in systemic risks are significant. Interestingly, India is relatively isolated from other country groups as its systemic risks are weakly correlated across other regions. Further univariate analysis shows that lead and lag effects in systemic risks are usually very short-term, at a quarterly level, and long-term effects fade out. Often, the highest correlation in systemic risks across markets is contemporaneous, implying that markets co-move with respect to underlying systemic risks. Granger Causality tests show that the systemic risks have strong momentum within each region, while the spillover effects of systemic risks across regions is weak, complementing similar findings for developed European countries in [Buch et al. \(2019\)](#) and [Sensoy et al. \(2017\)](#). Vector Auto Regression analysis confirms the earlier evidence that contemporaneous dependence of systemic risk across markets matters far more than lagged effects. It is therefore, unlikely that systemic risk in one of the emerging market regions leads that in another region.
3. A principal components analysis (PCA) of region-wide systemic risk measures shows that the first PC explains 52% of the variation in systemic risk, while the next two components explain about 20% of the variation each. The first three (four) components together explain 92% (97%) of the variation. The first PC correlates highly with the default risk during the financial crisis. The second PC spikes in the post-financial crisis period (associated with Dodd-Frank regulatory phase-in), reflecting possible policy uncertainty; and again during the foreign exchange crisis event of 2015-16. The third PC (i.e., PC3) seems to capture the taper tantrum of 2013 and the 2015-16 foreign exchange crisis, episodes both associated with capital outflows from emerging markets to US. Further analysis shows that changes in the first PC is significantly related to the US default factor. Changes in the third PC is weakly influenced by contemporaneous funding (TED).
4. Finally, we examine the information content of systemic risk in an out-of-sample setting. For all five geographic regions, we see that lagged changes in systemic risk are highly predictive of aggregate default risk (PD) in the following period. Interestingly, lagged PD is not. This suggests that our network measure of systemic risk provides explanatory power over and above the measure of credit risk levels in the economy. This suggests material ability to predict credit quality levels in economies using our new measure of systemic risk. This parallels a similar finding for developed European countries in [Pagano and Sedunov \(2016\)](#).

Systemic risk captures the conditional failure of the system at large, conditional on the failure of key financial institutions in an economy. Overall, in this project we undertake a large-scale study of systemic risk and credit cycles involving emerging market financial institutions in many countries and provide insights into policies for measuring, managing and regulating systemic risk. Our findings can be useful to academics, regulators, and financial practitioners.

The rest of the paper proceeds as follows. In [Section 2](#) we survey the now vast literature

on systemic risk and contagion in network models. Next, Section 3 undertakes an exploration of the comprehensive emerging market data, and reports some basic descriptive statistics. Our specific network construction methodology is explained in Section 4. Various network metrics are derived and estimated in Section 5. Section 6 conducts empirical analyses and section 7 concludes.

2 Literature

2.1 Definition and Origins

Systemic risk involves the risks that affect many market participants simultaneously, leading to severe losses, which then spread through the system. Systemic risk entails quick propagation of illiquidity and insolvency risks, and financial losses through the financial system as a whole, impacting the connections and interactions among financial stakeholders, especially so during periods of financial distress (Billio et al. (2012)), with potentially adverse consequences for the supply of credit to the real economy (Adrian and Brunnermeier (2016)). A broad literature suggests four possible sources of systemic risks viz., (i) banking related panics; (ii) banking crises arising from falling asset prices; (iii) contagion; and (iv) foreign exchange mismatches in the banking system (Allen and Carletti (2013); Brunnermeier and Pedersen (2009); Adrian and Shin (2010); Gorton and Metrick (2012); Acharya et al. (2013); Covitz et al. (2013))). Keeping these ideas in mind we develop a measure of systemic risk that is based on a network model and allows decomposition by individual financial institution.

2.2 Measuring Systemic risk

Surveys of systemic risk measurement include de Bandt and Hartmann (2000); Gale and Kariv (2007); Schwarcz (2008); Chan-Lau et al. (2016); Bisias et al. (2012); Benoit et al. (2017); Silva et al. (2017); and Anand et al. (2018). Two underlying approaches are used in the literature: (1) cross-sectional correlations, and (2) network-based measures.⁵

2.2.1 Cross-sectional Correlation Measures

In early work, Lehar (2005) uses a sample of international banks to estimate the dynamics and correlations between bank asset portfolios, where the asset portfolio for each bank is implied using the contingent claims model of Merton (1973). Huang et al. (2012) create the distressed insurance premium (DIP) measure, which captures systemic risk by calculating

⁵Other methods focus on data features other than correlations and networks, and deal mostly with tail risk measurement and principal components analyses, applied to non-equity markets, such as debt and CDS markets. These include systemic risk based on Contingent Claims Analysis (Saldías (2013)); LASSO methods (Demirer et al. (2017)); copula-based dynamic model (Oh and Patton (2018)); multivariate credit risk models (Li and Zinna (2014)); Bayesian methods (Bianchi et al. (2015)); tail event and extreme value driven models (Betz et al. (2016); Härdle et al. (2016); Li and Perez-Saiz (2018)); principal component or factor models (Avramidis and Pasiouras (2015); Nucera et al. (2016)); and vulnerability to fire-sale spillovers (Duarte and Eisenbach (2015)). Bianchi et al. (2019) propose a Markov Switching Graphical Seemingly Unrelated Regression (MS-GSUR) model to investigate time-varying systemic risk based on a range of multi-factor asset pricing models. Other papers present theoretical models for studying systemic risks e.g., He and Krishnamurthy (2013), and Dicks and Fulghieri (2019).

a hypothetical insurance premium against catastrophic losses in a portfolio of financial institutions. [Adrian and Brunnermeier \(2016\)](#) develop the conditional value at risk (CoVaR) model, which estimates the increase in the value at risk of the financial system conditional on a firm’s distress. [Acharya et al. \(2016\)](#) measure each financial institution’s systemic expected shortfall (SES), i.e., its propensity to be undercapitalized when the system as a whole is undercapitalized. While other measures of systemic risk measure the risk faced by the system as a whole, the SES metric measures the effect of systemic risk on an individual bank. [Brownlees and Engle \(2017\)](#) introduce the Conditional Capital Shortfall index for Systemic Risk Measurement (or SRISK) to measure the systemic risk contribution of a financial firm, which represents a bank’s expected capital shortfall when the market return is in the lowest 5% bracket in a given year, building on the marginal expected shortfall (MES) measure of [Acharya et al. \(2012\)](#). Compared to MES, SRISK incorporates information on a bank’s size and leverage. [Engle \(2018\)](#) examines the history and application of the SRISK measure. The above analyses relied on stock data but did not exploit network relationships, as we do in this paper.

2.2.2 Network-Based Measures

The Dodd-Frank Act stipulates that important banks are highly interconnected. [Nier et al. \(2007\)](#) investigate how systemic risk is affected by the structure of the financial system, i.e., interbank linkages. [Billio et al. \(2012\)](#) use return correlations and Granger causality regressions on returns to construct network maps and measures of systemic risk. [Billio et al. \(2012\)](#) apply several econometric measures of connectedness based on Granger-causality networks to the changes of sovereign risk of European countries. [Diebold and Yilmaz \(2014\)](#) provide several connectedness measures built from variance decompositions. [Elliott et al. \(2014\)](#) examine cascades in financial networks using a model of cross-holdings among organizations that allows for discontinuities in values. [Hautsch et al. \(2015\)](#) propose realized systemic risk beta as a measure of financial companies’ contribution to systemic risk, given network interdependence between firms tail risk exposures. [Markose et al. \(2012\)](#) study the network among CDS contracts to document the high concentration of exposures leading to a “too interconnected to fail” (TITF) phenomenon; [Ahern \(2013\)](#) finds that industries that are more central in the network of intersectoral trade have greater market risk and earn higher stock returns than industries that are less central, confirming the results of [Das and Sisk \(2005\)](#).⁶ Complementing these network models, our model of systemic risk networks provides a measure of systemic risk for the entire financial system, and each institution’s contribution to this risk, thereby providing an implementation pathway for measuring systemic risk, the identification and monitoring of systemically important financial institutions (SIFIs), and the effect on the level of aggregate credit risk in emerging market regions.

2.3 Managing Systemic risk

An effective systemic risk monitoring effort seeks to distinguish shocks, which are varied and difficult to predict, from vulnerabilities, which can amplify shocks and lead to instability ([Liang \(2013\)](#)). The regulatory framework in place prior to the global financial crisis was largely “microprudential” in nature, with the basic presumption that if each bank could be

⁶Other network risk papers include [Kitwattananachai \(2015\)](#), [Poledna et al. \(2015\)](#), [Bluhm and Krahnen \(2014\)](#), [Acemoglu et al. \(2015\)](#), [Brunetti et al. \(2015\)](#), [Donaldson and Micheler \(2018\)](#), [Anand et al. \(2013\)](#), [Chan-lau et al. \(2009\)](#), [Gabrieli and Georg \(2014\)](#), and [Colliard et al. \(2017\)](#).

prevented from taking large risks, there would not be a build-up of risk in the financial system. Post-crisis, regulation shifted towards a “macroprudential” approach, which recognizes the importance of general equilibrium effects, and seeks to safeguard the financial system as a whole (Hanson et al. (2011)).⁷ Our paper examines the macroprudential linkages between potential future default levels in the economy and current systemic risk metrics, which would aid policy makers greatly in managing the credit cycle better.

Kaserer and Klein (2019) find that a number of multi-line and life insurers exhibit levels of systemic risk similar to those of the riskiest banks in the sample. Our analysis, therefore, includes insurance companies as well as broker-dealers. Berger et al. (2019) find that government aid through Troubled Assets Relief Program (TARP) to banks significantly reduced contributions to systemic risk, particularly for larger and safer banks, and those in better local economies, by increasing the value of common equity and increasing capital cushion. Our analysis, which shows that systemic risk is concentrated in a small number of major banks, and the primary principal component of systemic risk is default levels, also supports similar policy prescriptions, consistent with Acharya et al. (2010), who suggest that a risk-based deposit insurance premium should not only reflect the actuarial fair value but should also include an additional fee imposed on SIFIs to reflect their excessive risk taking and the disproportionate cost they impose on others in the system.⁸

Another principal component of systemic risk in our empirical analysis is liquidity risk. Brunnermeier and Pedersen (2009) emphasize the usefulness of a capital surcharge to reduce liquidity risk associated with maturity mismatches, while Perotti and Suarez (2009) propose a mandatory tax on wholesale funding that could be used to fund an insurance scheme. Goodhart (2009) proposes a liquidity insurance mechanism in which access to publicly provided contingent liquidity would be permitted if a premium, tax, or fee were paid in advance.

2.4 Systemic Risks in Developed Countries

A plethora of empirical studies exist on systemic risk in developed economies. These include⁹ (a) Li and Zinna (2014), who show that systemic risks in the US and UK differ both in their evolution, and in their banks’ systemic exposures; (b) Giglio et al. (2016) study how systemic risk and financial market distress affect the distribution of shocks to real economic activity; (c) Sedunov (2016) finds that CoVaR provides better forecasts of future systemic risk exposures compared to systemic expected shortfall (SES) and Granger causality; (d) Laeven et al. (2016) examine cross sectional variation of standalone and systemic risks of large banks during the global financial crisis; (e) Pagano and Sedunov (2016) show that the aggregate systemic risk exposure of financial institutions is positively related to sovereign debt yields in European countries; (f) Bostandzic and Weiß (2018) provide evidence that, compared to US banks, European banks appear to contribute more to global systemic risk, because of the lower quality of their loan portfolios and their higher relative interconnectedness with the rest of the global financial system; (g) Karolyi et al. (2018) find that cross-border

⁷The Dodd-Frank Act (2010) promoted a macroprudential approach to supervision and regulation. The Act suggested the designation of financial institutions as “systemically important” and such institutions are required to maintain additional regulatory risk buffers (about 1% additional capital).

⁸Other related work on systemic risk regulation includes Gobat et al. (2011), Abbass et al. (2016), Benoit et al. (2019), Löffler and Raupach (2018), Roukny et al. (2018) and Buch et al. (2019).

⁹See also Tasca et al. (2014); Liu et al. (2017); Black et al. (2016); Varotto and Zhao (2018); Aparna Gupta et al. (2018); Adrian and Boyarchenko (2018); and Ellul et al. (2018).

bank flows reduce systemic risk by improving banks asset quality, efficiency, and reliance on nontraditional revenue sources; (h) [Cai et al. \(2018\)](#) show that while loan syndication improves institution-level risk reduction through diversification, it can induce higher bank interconnectedness; (i) [Albuquerque et al. \(2019\)](#) show that incentive pay based on relative performance evaluation can lead managers disproportionately to choose investments that are correlated across firms, thus increasing systemic risk; (j) [Anderson et al. \(2019\)](#) examine how National Banking Acts (NBAs) of 1863–1864 changed the network structure and affected systemic risk; and (k) [Das et al. \(2018\)](#) study how bank failures altered the network of financial institutions during the Great Depression. We find parallels to many of these results in emerging markets.

2.5 Systemic risks in Emerging markets

There is a limited prior literature on the evidence of systemic risk in emerging markets. For example, [Sensoy et al. \(2017\)](#) find that the increased average correlation among emerging market sovereign bond returns is more likely caused by clusters of countries that exhibit high “within-cluster” co-movement rather than “between-cluster” co-movement. [Sensoy \(2017\)](#) finds that institutional ownership in Turkey leads to an enhanced systematic liquidity risk for mid-to-large cap firms by increasing the commonality in liquidity. [Borri \(2018\)](#) adopts the CoVaR risk measure to estimate the vulnerability of emerging market countries to systemic risk in the market for local currency government debt. [Fang et al. \(2018\)](#) show that idiosyncratic risk of Chinese financial institutions is significantly affected by prevalent risk networks. [Wang et al. \(2018\)](#) investigate the interconnectedness and systemic risk of China’s financial institutions and find that large commercial banks and insurers usually exhibit systemic importance, but some small firms are systemically important due to their high level of incoming (outgoing) connectedness. While these papers focus on one or a few emerging market countries, our paper covers a bigger sample of 1048 FIs in 23 countries across 5 regions.

3 Data

We first identify the list of 23 emerging countries by combining the IMF’s and MSCI’s lists of emerging countries and further intersecting it with the CDS data available in the Markit database. Intersecting with CDS data helps us preserve only those firms where the public debt outstanding is sizable and there is market-wide exposure to the underlying credit risk. Using Datastream, we extract a comprehensive list of financial firms from these 23 markets. We require that our sample consists of active financial firms, and firms whose common equity are major securities trading on a primary exchange in the local market. We exclude (a) non-financial firms, (b) inactive (delisted) firms, (c) firms with only preferred stock, (d) foreign firms, and (e) firms trading exclusively in either a minor exchange in the local emerging market or a foreign exchange. We also drop firms with less than 125 active trading days (or six calendar months of exchange history).

Based on International Securities Identification Number (ISIN) and/or Stock Exchange Daily Official List (SEDOL) identifiers, we further match emerging market financial firms to the Compustat Global Database and obtain the corresponding GVKEYs and Standard Industrial Classification (SIC) codes. Based on SIC codes, we categorize firms as (a) Banks

(SIC: 6000-6199), (b) Broker-Dealers (SIC: 6200-6299), (c) Insurers (SIC: 6300-6499), and (d) Others (all other SICs). We eliminate firms with no SIC code and firms classified as others (which include financial subsidiaries of non-financial corporations and specialized investment vehicles such as funds, REITs and securitized assets). Table 1 presents our final sample of emerging market financial firms. The final screened sample consists of 1048 financial institutions, comprised of 539 Banks, 389 Broker-Dealers and 120 Insurers. Overall India accounts for the highest proportion of the sample with 387 financial firms or 37% of the total sample, followed by Indonesia (8%) and China (6%). There is a high clustering of the sample among top five countries (i.e., India, Indonesia, China, Poland, and Thailand), which together account for 62% of the whole sample.

[Insert Table 1 here]

Using Datastream, we extract daily equity returns—both dividend- and stock-split-adjusted—spanning a 13 year period from 2004 to 2016. We linearly interpolate any missing daily returns. In addition, based on ISINs and/or SEDOLs, we obtain distances-to-default (DTD) and probabilities of default (PD) for 7 maturities: 1, 3, 6, 12, 24, 36 and 60 months, from the Credit Research Initiative (CRI) Database maintained at the Risk Management Institute (RMI) of the National University of Singapore (NUS). The database reports monthly DTD and PD values computed from Merton-type models using firm-specific values; these monthly values are converted into daily time-series corresponding to returns.

We also collect several balance sheet and income statement variables corresponding to the financial institutions from Datastream on a quarterly basis and compute the following firm-specific quarterly attributes: (i) Log(Assets) and Log(Market Cap) as measures of firm size in terms of book value of assets and market value of equity, respectively; (ii) Loans/Assets and Loans/Deposits ratios to capture banks' focus on traditional lending activities and core financing activities (these ratios are set to zero for non-bank financial institutions); (iii) Debt/Assets and Debt/Equity ratios to capture leverage; (iv) Debt/Capital as a measure of the liquidity position of the financial firm; (v) ROA (return on assets) and ROE (return on equity) as measures of operating performance of the financial firm; and (vi) Market/Book value of equity ratio of the financial institution as a measure of the stock price based performance.

4 Network Construction

We use the return data to construct networks using a modified Granger causality approach. Our approach is an extension of the method in Billio et al. (2012). In their original method, for any two banks i, j , we run the following regression:

$$r_{j,t} = a + b \cdot r_{j,t-1} + c \cdot r_{i,t-1} + e_{j,t} \quad (1)$$

where $r_{i,t}$ denotes return for bank i on day t . If coefficient c is significant (we use a p -value less than 0.025), then we assign a network link from bank i to bank j . This means that if bank i experiences a shock it will transmit the shock to bank j . Likewise, we can run the reverse regression to determine if a risk transmission link exists from bank j to bank i . We run pairwise regressions for all banks, i.e., for n banks we have $n(n-1)$ regressions. We store the network links in a network adjacency matrix denoted A of size $n \times n$. Here, $A(i, j) = 1$ if there is a risk spillover from bank i to j , else $A(i, j) = 0$; and $A(i, i) = 0$.

This approach has been criticized as both banks may have co-movement on account of a joint factor, i.e., the returns on an overall index of FIs, see [Chan-Lau et al. \(2016\)](#) for a survey and critiques of network construction models. To exclude this effect and focus only on the pure linkage between two banks, we modify the regression above to include lagged values of the equal-weighted return ($r_{EW,t-1}$) of all banks used to construct the network. This variable soaks up any lagged co-movement, thereby isolating the idiosyncratic risk spillover between two banks. Our new specification is as follows.

$$r_{j,t} = a + b \cdot r_{j,t-1} + c \cdot r_{i,t-1} + d \cdot r_{EW,t-1} + e_{j,t} \quad (2)$$

where $R_{EW,t}$ is the equal-weighted return of all banks for day t . Again, to establish the link $A(i, j) = 1$, we require that the p -value of the coefficient c be small, i.e., $p \leq 0.025$, if $c > 0$. Note that if $c \leq 0$, then there is no risk spillover from i to j , in which case we also set $A(i, j) = 0$.

To construct the network matrix for any day t , we have to make choices about the look-back period of returns, and which banks to include in the analysis. These choices are as follows.

1. The look back period is chosen to be $L = 130$ trading days, i.e., roughly a half-year.
2. For the chosen period, we extract all bank returns, and exclude any bank that does not exist through the entire period.
3. For the remaining banks, we find that many banks have stock prices that do not move much, and are illiquid. These are essentially very small banks that are not likely to have any systemic effects. If stock prices remain same from day to day, returns will be exactly zero on many days. Therefore, we exclude all such banks that have zero returns on more than $1/3$ of the sample L days.
4. We then run the network construction model described above to create the adjacency matrix A . We do this for each quarter end starting with Q3 2004, ending with Q4 2016. This provides a total of 50 quarters, and a network for each one.

5 Network Statistics

In order to detect which nodes are most influential in the network, we compute betweenness centrality from the adjacency matrix A .¹⁰ The definition of betweenness centrality for node

¹⁰An alternate measure is “eigenvalue centrality,” originally defined in [Bonacich \(1987\)](#), and further discussed in [Bonacich and Lloyd \(2001\)](#), defines centrality of a node i to be a linear function of the connection strength A_{ij} with nodes j it is connected to, and the centrality of those nodes, c_j . This leads to a circular system of simultaneous equations:

$$c_i = \sum_{j=1}^n A_{ij} c_j, \quad \forall i \quad (3)$$

One solution to this system of equations is the principal eigenvector in an eigenvalue decomposition of matrix A . This vector contains n components $c_i, i = 1, 2, \dots, n$.

v is as follows, see [Freeman \(1977\)](#):

$$b_v = \sum_{\substack{i,j \\ i \neq j \\ i \neq v \\ j \neq v}} \left[\frac{g_{ivj}}{g_{ij}} \right] \quad (4)$$

where g_{ivj} is the number of shortest paths from node i to node j that pass through node v , and g_{ij} is the number of shortest paths from i to j . We preferred to use betweenness centrality as it is more robust when there are many nodes in the network that are unconnected, in which case eigenvalue decomposition required for computing eigenvalue centrality becomes less stable. Further, betweenness centrality directly picks up the nodes through which risk passes fastest, since these nodes sit on the shortest paths between other nodes, and will facilitate transmission of risk spillovers. However, for a well connected network, eigenvalue centrality may provide a better depiction of the importance of each node.

5.1 Network Metrics

There are several statistics that we compute from the adjacency matrix representing the bank network. These are as follows. First, we have two bank-level metrics:

1. Degree: the number of connections of each node, which characterizes how interconnected the network is. The degree distribution also reveals how concentrated the network connections may be in a few nodes, as often occurs in hub and spoke networks.
2. Betweenness centrality. A measure of how central position a bank has in the network for a given bank. A node is said to be “between” other nodes when a large proportion of shortest paths in the network pass through the particular node.

In addition, we have seven aggregate network-level metrics, as follows:

1. The number of nodes. The larger the number of nodes the greater the possible connectivity and transmission of network risk, and possibly greater systemic risk.
2. Diameter. We calculate clusters, i.e., groups of connected nodes, and diameter is defined as the longest shortest path between any two nodes in the largest cluster in the network. It is a measure of how much time it would take for a problem at one side of the network to reach the opposite side. It is one quantification of risk transmission. Networks with a small diameter are more likely to experience system-wide problems.
3. Mean degree. We calculate $E(d)$, where d_i is number of connections of node i in the network. Mean degree depicts the average number of links each node in the network has.
4. Fragility. We define the fragility score of the network as $E(d^2)/E(d)$. The numerator is the raw Herfindahl index of the degree distribution, and is higher if connections are concentrated in a few nodes. The denominator normalizes this score by dividing by mean degree. The higher the fragility or concentration in the network, the greater

is the likelihood that a local problem in the banking network will spread across the network and become a global problem. High fragility is a property of hub and spoke networks. Once a problem reaches and impacts a hub node, it then spreads rapidly through the network.

5. Herfindahl index of degree (Degree HHI). The Herfindahl index is calculated as $H = \sum_{i=1}^n \left(\frac{d_i}{\sum_{i=1}^n d_i} \right)^2$. We normalize it so as to get a value between 0 and 1, by computing $NH = \frac{H-1/n}{1-1/n}$.
6. Clusters: we also examine how disconnected the network is by computing the number of connected groups, i.e., clusters, where a cluster is an independent group of nodes that is not connected to any other group of nodes. The greater the number of disconnected clusters, the more insulated the network is from economic contagion.
7. Herfindahl index of cluster sizes (Cluster HHI): This is a measure of the concentration in nodes, and measures if nodes are resident in one large cluster or if the clusters are balanced in size.

We calculate all the nine network statistics for all quarter end dates starting from Q3 2004 to Q4 2016, a total of 50 quarters.

We next examine the evolution of systemic risk across countries over time. Table 2 presents the network measures across geographic regions for three periods centered around the crisis, i.e., pre-crisis, crisis, and non-crisis periods that encompass the years 2004-2006, 2007-2009, and 2010-2016 respectively. We consider 23 countries clustered into five geographical regions: East Asia (China, Indonesia, Malaysia, Philippines, South Korea, Taiwan, Thailand), South Asia (India), Eastern Europe (Bulgaria, Czech, Hungary, Poland, Russia, Ukraine), Southern Europe and Africa (Egypt, Greece, South Africa, Turkey) and South America (Argentina, Brazil, Chile, Columbia, Mexico). We report average values of network measures aggregated at the country/regional level and the firm/regional level. Both East Asia and South Asia have the highest systemic risks based on nodal level (number of nodes, mean degree, between centrality) and network level (fragility, number and Herfindahl index of clusters) risk measures. Most of the risks are concentrated during crisis and post-crisis periods. Our sample of the banking networks is larger for Asia and smaller for Europe, Africa, and South America. Mean degree of a node in the network is also higher in Asia (> 7.3) versus the other regions (ranging from $> 1.5 - 3.0$). Correspondingly, the betweenness centrality is much higher as well because it is not normalized. The diameter of the bigger networks is also greater as is to be expected. Because the Asian networks are bigger and more interconnected, the fragility is higher and the number of independent clusters is fewer. Furthermore, South America has high network level risks based on diameter and Herfindahl index of nodes measures.

Network connections are based on the Adjacency matrix populated using Granger regressions using p-values of 0.025, as discussed in Section 4. All average values reported are based on metrics computed every quarter. Table 2 shows that network risks have increased during the crisis based on fragility and the Herfindahl index of degree for all the regions. Other metrics such as the Herfindahl index of clusters, number of nodes, and betweenness centrality indicate a higher network risk for the post-crisis period across regions.

[Insert Table 2 here]

5.2 Risk Metrics

The network adjacency matrix A describes the structure of risk spillovers between banks. But the network does not account for the total potential impact of these risk spillovers on the system as a whole, i.e., systemic risk. We therefore, combine network information with credit information using the systemic risk score developed in Das (2016); Das et al. (2019). We deploy a modified version of the risk score in prior work by weighting credit risk by bank size, and converting it into a risk score per bank instead. This normalizes the score so that we may proceed to use it for comparison of systemic risk across time, even as the number of banks changes. Our region-wide scalar measure, denoted S , is as follows:

$$S = \frac{1}{n} \sqrt{C^\top \cdot A \cdot C} \quad (5)$$

where n , as before, is the number of banks, and $C = a \cdot \lambda$ is a n -vector of size-weighted credit risk scores of each bank where $a = \log(\text{TotalAssets})$ and λ is a credit quality measure. We require that λ be increasing in credit risk. We make the following observations.

1. Before computing S , the adjacency matrix A is further updated to set the diagonal to be 1. In the special case where no bank is connected to another, i.e., $A = I$ (where I is the identity matrix), systemic risk is simply the L2 norm of credit risk vector C , i.e., $\frac{1}{n} \|C\| = \frac{1}{n} \sqrt{C \cdot C}$.
2. There are many conceivable ways to construct the λ vector. Examples are credit ratings converted into integer scores, with rating AAA = 1, AA = 2, etc. We may also use probability of default (PD), the reciprocal of distance-to-default, or a sparse scoring system where investment grade and below-investment grade are given a lower and higher chosen values. In this paper, we employ PDs as a default proxy to obtained our default weighted systemic risk measure.
3. Because we normalized the score by n , we may compare this score across countries, and across epochs for the same country. The S score represents a per-bank, dollar-weighted, and network-weighted credit risk measure for the entire financial system.
4. Noting that all elements of A are positive, i.e., $A_{i,j} \in \{0, 1\}, \forall i, j$, and that $C_i \geq 0, \forall i$, systemic risk is non-negative, i.e., score $S \geq 0$.
5. An increase in any element of A (network effect) or C (individual risk effect) will result in an increase in S .
6. The function $S(C, A)$ is linear homogenous in C . Using this property, and applying Euler's homogeneous function theorem¹¹, we see that

$$S = \frac{\partial S}{\partial C_1} C_1 + \frac{\partial S}{\partial C_2} C_2 + \dots + \frac{\partial S}{\partial C_n} C_n = \sum_{i=1}^n \frac{\partial S}{\partial C_i} C_i \quad (6)$$

and each component $\frac{\partial S}{\partial C_i} C_i$ of this equation comprises the “risk contribution” of bank i to total systemic risk. This allows a regulator to apportion systemic risk to each bank such that it is additive across all banks.

¹¹<http://mathworld.wolfram.com/EulersHomogeneousFunctionTheorem.html>.

7. The expression $\frac{\partial S}{\partial C_i}$ in closed-form is as follows:

$$\frac{\partial S}{\partial C} = \frac{1}{2n^2 S} [A \cdot C + A^\top \cdot C] \in \mathcal{R}^n \quad (7)$$

which provides the entire vector in one matrix calculation making for efficient computation. Therefore, S may be written as the following scalar quantity:

$$S = \frac{1}{2n^2 S} ([A \cdot C + A^\top \cdot C] \odot C)^\top \cdot \mathbf{1} \quad (8)$$

where $\odot$ stands for the Hadamard product of two vectors or matrices, and $\mathbf{1}$ is a n -vector of 1s, i.e., a unit n -vector. And clearly, the risk contribution of any bank i is a consequence of equation (6):

$$\frac{\partial S}{\partial C_i} \cdot C_i = \left\{ \frac{1}{2n^2 S} \cdot [A \cdot C + A^\top \cdot C] \odot C \right\}_i, \quad \forall i. \quad (9)$$

We employ these various metrics in the following empirical analyses.

6 Empirical Analyses

6.1 Distribution of the Probability of Default

We present the average quarterly values of probability of default, PD, and systemic risk score measures over time for five geographical regions. We use the one-year PDs in our analysis as is commonly done in the credit risk industry. In order to create the vector C that we need to compute systemic risk, we map the PDs into C as follows:

$$C = \lambda \cdot \ln(A) \quad (10)$$

where A is total assets of the bank. $\lambda = (1 + 30PD)$ is a scaled value of PD, where the scaling is applied only to increase the size of the value of λ because PD values are extremely small. For all the banks included in the data set each quarter, we calculated the systemic risk score S , using the C vector as noted earlier. For each quarter the element of the C vector is computed using the mean PD for each bank across the days in that quarter. If there are a few days data of PD missing in the quarter, then the mean is calculated over the data on days for which it is available. In the rare case when a bank has no PD data for any days in the quarter, we ascribe the bank's C value to the mean PD across all the other banks in the sample for that quarter. Figure 1 plots the time series of PD for the entire sample period from 2004 to 2016, for each of the five geographic regions. Panel A presents levels while Panel B shows the changes in PD over time. Panel A shows a wide variation in regional PD levels over time. PDs from all regions go up during the financial crisis period. As is evident from both panels, the East European PD experiences the largest spike during the crisis. South Asian PD registers a large increase during the 2013 taper tantrum period, and also during the 2015 exchange rate crisis. We also see an increase in regional PDs during the 2011-2012 European debt crisis.

[Insert Figure 1 here]

The time series of systemic risk is shown in Figure 2. The systemic risk metric (equation (5)) is normalized for the number of banks, which has been increasing over time. In general, systemic risk spikes during the financial crisis period. The three regions East Asia, Eastern Europe, and South America, experience large increases in systemic risk compared to other regions over time mainly during the 2009 crisis. South Asia (and to a lesser extent, Southern Europe & Africa) registers its highest systemic risk during the 2013 Taper tantrum period and again during the 2015-16 foreign exchange crisis period. We also observe that the evolution of PD and systemic risk changes do not perfectly mirror each other.

[Insert Figure 2 here]

Table 3 presents the average quarterly values of PD and systemic risk over time for the five geographical regions. We find that South Asia has the highest mean PD; PD is also high for Eastern Europe and Southern Europe & Africa, more so during the crisis period. Systemic risk per bank (measured as number of nodes) is significantly higher in East Asia and South Asia relative to the other three regions. Systemic risk is also reported at the aggregate network-level. Systemic risk is high at a network level for the full sample and pre-crisis period for Eastern Europe and South America compared to other regions. In summary, we see differences in regions that experience a rise in both firm- and network-level systemic risks. Systemic risk and PD movements do not always line up, suggesting that network risk based on the pattern of inter-connectedness also plays a key role.

[Insert Table 3 here]

We examine whether there is persistence in the time series of systemic risk, in levels or in changes. The results are shown in Figure 3. In the autocorrelation plot in levels, there is clear evidence of autocorrelation to at least one quarter lag in most regions, though in the case of two of the regions—Eastern Europe and South America—there is greater persistence. The auto-correlograms for changes in systemic risk show much less persistence. Again, however, Eastern Europe shows some positive autocorrelation in changes at one lag whereas South America evidences negative autocorrelation at one lag. We will explore the use of these time series in predicting aggregate levels of default risk later in the paper.

[Insert Figure 3 here]

Does most of the systemic risk come from just a few banks? To investigate this, we apply equation (9) to compute the percentage of systemic risk contributed by the top 10 contributors in the full sample and each subperiod. This is shown in Table 4. Panel A reports the average contribution, as a percentage of total regional systemic risk, by the top 10 contributing institutions of each region using the data from 2004-Q1 to 2016-Q4. We observe that top 10 percentile contributors systemic risk explains 16% to 47% of the systemic variation. Eastern Europe and South America have the maximum concentration (over 46%) of systemic risk among the top 10 contributors. We also observe that the percentage contribution for each geographic region in general is much higher in the pre-crisis period. Panel B reports the average number of contributing institutions that account for the top 50 percentile (that is, contribute at least 50 percent) of total regional systemic risk. Eastern Europe and South America has the smallest number of banks (i.e. 12 and 11 respectively)

in the top 50 percentile of the contributors of systemic risk. In general, most of the systemic risk for each geographic region is concentrated among fewer banks in the pre-crisis and crisis periods compared to post-crisis period. There is a clear bifurcation here: East Asia and South Asia have less concentration of systemic risk in their top 10 banks than do the other three regions. This is clearly an outcome of their bigger banking networks that enable greater spreading of systemic risk.

[Insert Table 4 here]

6.2 Time-series and cross-sectional regressions of quarterly regional systemic risk

We seek to explain the evolution of aggregate network-level systemic risk over time. The objective here is to examine the key drivers of systemic risks across regions. To this end, we conduct time-series regressions (over 50 quarters from the third quarter of 2004 through the fourth quarter of 2016) of network systemic risk changes on several covariates: (a) aggregate credit risk (mean probability of default across firms), (b) various network parameters (mean degree across all nodes, degree concentration measured by HHI, mean centrality between nodes, diameter, fragility, number of distinct clusters, and HHI concentration within clusters), and (c) median firm-specific attributes (book value of assets, market value of equity, loans-to-assets and loans-to-deposits ratios of banks, debt-to-assets and debt-to-equity ratios, debt-to-capital ratio, returns on assets and equity, and market-to-book value of equity); and (d) U.S. macroeconomic variables (default factor, level and slope of term structure factor, TED spread, and VIX). All regressions are implemented in first-differences and include adjustments for heteroskedasticity, autocorrelation in residuals, and year fixed effects.

Table 5 presents the time series regressions for the five geographic regions respectively in Panels A to E. Taking East Asia (Panel A) as an example, we find that the variables that quantify credit risk explains about 47% of the variation in changes in systemic risk (regression 1). Node-level aggregate metrics explain 87% of this variation (regression 2). Network-level variables alone give an explanatory power of 86% (regression 3). Putting all these sets of variables together explains 94% of systemic risk variation (regression 4). Prevailing firm-specific attributes in aggregate add another 3% and in general provide very little explanation of the evolution of systemic risk over time (regressions 5 and 6). Adding the US based default, term structure, and risk aversion factors increases the explanatory power additionally by 2% to a total of 98% (regressions 7 and 8). Taken together, we can explain almost 98% of the variation in systemic risk over time for East Asia. Table 7, Panel A summarizes the results of the regressions for East Asia and other regions. We observe from Table 5 (Panels A to E) and Table 7, Panel A that overall across all five regions, credit risk explains a relatively lower portion of systemic risks in Southern Europe & Africa (32%) and South America (0%); network risk at the same time explains the highest portion of systemic risks in East Asia (86%–87%) and South America (90%–93%). In addition, the contribution of firm-specific attributes and US macro factors accounts for 16% (for South Asia) and 0% (Southern Europe & Africa) of the explanatory power. Besides mean PD, we observe that several systemic variables are significant (mean degree, degree HHI, diameter, fragility, number of clusters, and cluster HHI) across regions. As a robustness check, we also re-ran the entire network construction using Granger regressions where the confidence level for significance of the link coefficient is taken to be 0.99 instead of 0.975 (these results are omitted for brevity); we find that the structure and fit of the network model is similar.

[Insert Tables 5 and 7 here]

Next we consider panel regressions of quarterly systemic risk contributions of all emerging market banks. The regressions are once again implemented in first differences. Panel A, Table 6 reports the results for all geographical regions considered jointly, and Panels B through F present the results for individual geographical regions. The dependent variable is the changes in each firm’s contribution to network level systemic risk score. (Panel A regressions include region-specific cluster effects. All panel regressions include year fixed-effects, and adjustments for heteroskedasticity.) We find that the overall credit risk explains about 9% of the variation in individual firms’ systemic risk (regression 1). Adding network risks increases the explanatory power substantially to 64%. Firm-specific attributes add another 4%. Table 6 (Panels B to F) and Table 7, Panel B show that across all five regions, credit risk explains a relatively lower portion of systemic risks in East Asia (21%) and South America (21%). At the same time, network risk explains the highest portion of systemic risks in these regions: Eastern Europe (41%-64%) and East Asia (8%-61%). In addition, the contribution of firm-specific attributes is highest for Eastern Europe (15%) and South America (19%), while the US macro-factor contribution is negligible. Cross-sectional variation of firm-specific attributes matters in explaining systemic risks as reflected in the panel regressions. Besides mean PD, we observe that several systemic variables are significant (mean degree, degree HHI, diameter, fragility, number of clusters, and cluster HHI) across regions. Collectively, time series and panel data regressions imply that changes in PD and network risk variables together explain much of systemic risk changes. There is a cross-sectional heterogeneity across regions. Table 7 shows that role of credit risk is generally lower for South America (both time series and panel regressions) and East Asia (panel regression); similarly the role of network risks is lower for South Asia (time series regression) and Southern Europe & Africa (panel regression).

[Insert Table 6 here]

The systemic risk measure S is a combination of interconnectedness and default risk. Our empirical assessment of the role of each component suggests that interconnectedness plays an important role as it explains more of the changes in S than default risk.

6.3 Analyzing cross-market information flows of systemic risks across regions

We next analyze how systemic risks are connected across regions. We employ cross-correlation, Granger causality, and VAR tests to examine how systemic risks flow across regions. The objective is to better understand the nature of cross-market lead-lag relationships among systemic risks.

We first measure correlations of systemic risk between regional groups, where systemic risk is obtained as the regional network level systemic risk score. Table 8, Panel A reports contemporaneous quarterly correlations, while Panel B reports one-quarter lagged correlations. Panel A shows that contemporaneous correlations are significantly high between East Asia and three non-Asian regional blocks (i.e. Eastern Europe, Southern Europe & Africa, and South America). Eastern Europe and South America show strong contemporaneous correlations. Interestingly, South Asia (consisting of India), has no significant correlations

with any other region, and hence, is relatively isolated. Panel B shows that significant lagged correlations exist across markets. One quarter lagged systemic risk from East Asia significantly influences systemic risks in Eastern Europe and South America. Similarly lagged systemic risk from Southern Europe and Africa significantly affects South Asian systemic risks. Overall, we find that both contemporaneous and lagged correlations matter in the evolution of systemic risks across regions. South Asia is again isolated from other country groups as lagged correlations are very small and trivial versus the other four groups.

[Insert Table 8 here]

We next present pairwise cross-correlograms across markets. Figure 4 presents the corresponding plots. We consider pairs of geographic regions and study lead and lag relationships between their respective systemic risks. In each plot, to the right of zero ($x\text{-axis} > 0$), the first-named group leads the second-named group; similarly, to the left of zero ($x\text{-axis} < 0$), the first-named group lags the second-named group. Specifically, in the first plot, East Asia negatively leads South Asia ($x\text{-axis} > 0$) in the short-term over five quarters, and positively lags South Asia ($x\text{-axis} < 0$) in the long term over five years. Similarly, we find lead lag effects across many markets.¹² Generally, we observe that lead and lag effects are usually very short-term. Long-term effects fade out. Often the highest correlation is contemporaneous ($x\text{-axis} = 0$). The second plot in Figure 4 shows the cross-correlograms for quarterly changes in systemic risks across regions. Lead-lag effects are not noticeable.

Therefore, to better understand the auto- and cross-correlation relationships across regions, we implement Granger causality regressions. For each country regional group, the quarterly systemic risk measure is regressed on one-quarter lagged values of systemic risk measures of all five regional groups (including itself). Table 9, Panel A reports the univariate F-statistics of significance. Panel B reports the joint F-statistics of significance for the four other cross-regional groups considered together. We observe that dependence on self-lagged variables is usually strong (the diagonal terms) but dependence on cross-lagged variables is usually weak. We find cross-market lead-lag evidence only across three blocks: (a) East Asia significantly Granger causes the systemic risks in Eastern Europe, Southern Europe/Africa and South America. Similarly, (b) Eastern Europe significantly Granger causes the systemic risks in Southern Europe/Africa, and finally (c) Southern Europe/Africa significantly Granger causes the systemic risks in South Asia. Overall, the results in Table 9, Panel A, show that lagged effects are significant in select markets. Joint Granger causality tests from Panel B show that systemic risks in Eastern Europe, Southern Europe & Africa, and South America significantly depend on joint cross-lagged variables from other markets.

[Insert Figure 4 and Table 9 here]

To better understand the linear time-series interdependencies between the systemic risks we implement the following Vector Auto-Regressions (VAR) model across the five country groups.

¹²East Asia positively leads and lags Eastern Europe and South America. East Asia positively (negatively) leads (lags) Southern Europe & Africa. Eastern Europe positively leads and lags South America. South Asia negatively lags Eastern Europe, Southern Europe & Africa, and South America.

$$\begin{pmatrix} SysriskEastAsia \\ SysriskSouthAsia \\ SysriskEastEurope \\ SysriskEuropeandAfrica \\ SysriskSouthAmerica \end{pmatrix}_t = \text{intercept} + \begin{pmatrix} SysriskEastAsia \\ SysriskSouthAsia \\ SysriskEastEurope \\ SysriskEuropeandAfrica \\ SysriskSouthAmerica \end{pmatrix}_{t-1} \\
+ \dots + \begin{pmatrix} SysriskEastAsia \\ SysriskSouthAsia \\ SysriskEastEurope \\ SysriskEuropeandAfrica \\ SysriskSouthAmerica \end{pmatrix}_{t-4} + error_t$$

The VAR model involves quarterly network level scores of systemic risks that are jointly regressed on four lagged (one through four quarters) values of systemic risk measures of the five regional groups. Both likelihood ratio (LR) and Akaike information criterion (AIC) identify that a maximum of four lags are material. Out of 100 explanatory variables (5 regressions $\times$ 5 systemic risk scores $\times$ 4 lags), only eight coefficients are significant. This implies that consistent with all other results across country groups, contemporaneous dependence of systemic risk matters far more than lagged inter-dependence. Table 10 reports the summary of coefficients and t-statistics (corresponding p-values in parentheses) of eight such regressions. Out of the eight regressions, four regressions have significant 1-quarter lags, two have 2-quarter lags, and one each for 3- and 4-quarter lags. Similarly, two out of the eight regressions have self-lag dependence while six others have significant cross-lag dependence.

[Insert Table 10 here]

In summary, our findings in this section imply that systemic risk changes are strongly autocorrelated within each region and to a lesser extent, contemporaneously correlated across regions. Spillover effects across regions show up in univariate tests, though tend to be less prominent in Granger causality and VAR regressions.

6.4 Analyzing systemic risks using principal components

We further examine the key economic drivers behind systemic risks. We undertake a principal components analysis (PCA) of region-wide systemic risk measures. Table 11, Panel A presents the first five PCs and corresponding eigenvalues for the regional network level systemic risk score of the five regional country groups. We observe that the first PC explains 52% of the variation, while the next two components explain about 20% of the variation each. The first three (four) components together explain 92% (97%) of the joint variation in the regional time series of systemic risk. Panel B further shows that first three PC changes together account for 82% of the variation in systemic risk changes.

[Insert Table 11 here]

Figure 5 plots the first three PCs (levels and changes). We observe that the first PC (i.e., PC1) correlates highly with default risk during the financial crisis. PC2 spikes in the

post-financial crisis period (associated with the Dodd-Frank regulatory phase-in), reflecting possible policy uncertainty shock; and again during the foreign exchange crisis event of 2015-16. The third PC (i.e. PC3) seems to capture the taper tantrum of 2013 and the 2015-2016 foreign exchange crisis, episodes both associated with capital outflows from emerging markets to US. Figure 5 also plots the PC changes extracted from regional-level systemic risks. The changes in the first PC captures the credit risk changes from the financial crisis. The second and third PCs, however, are less distinct and broadly capture the 2013 taper-tantrum and 2015 foreign exchange crisis.

[Insert Figure 5 here]

We next study how the PCs are related to the underlying macro factors. Table 11, Panel C reports the results of quarterly time-series regressions of the U.S. macroeconomic variables (default factor, level and slope of term structure factor, TED spread, and VIX). Regressions include adjustments for heteroskedasticity, autocorrelation, and year-fixed effects. We observe that the first PC is significantly related to the US default factor. The third PC is influenced by the contemporaneous funding (TED) factor. The second PC does not appear to be related to any of the macroeconomic variables.

6.5 Out-of-sample prediction of aggregate default risk

Finally, we examine the information content of systemic risk in an out-of-sample setting. We consider the predictability of default risk using lagged systemic risk measures. We consider predictive time-series regressions of quarterly changes in credit risk on lagged values of explanatory variables. Table 12, Panels A through E report the results for individual geographical regions. The dependent variable is the change in credit risk (mean PD) between quarters t and $t - 1$. Explanatory variables include: quarterly changes in probability of default; quarterly changes in systemic risk (network level systemic risk score); quarterly changes in network attributes (mean degree across all nodes, degree concentration measured by HHI, mean centrality between nodes, diameter, fragility, number of distinct clusters, and HHI concentration within clusters); quarterly changes in median firm-specific attributes (book value of assets, market value of equity, loans-to-assets and loans-to-deposits ratios of banks, debt-to-assets and debt-to-equity ratios, debt-to-capital ratio, returns on assets and equity, and market-to-book value of equity); and quarterly changes in U.S. macroeconomic variables (default factor, level and slope of term structure factor, TED spread, and VIX). Mean PD, degree and centrality across firms, and market-wide median firm-specific attributes are computed every quarter. All regressions include adjustments for heteroskedasticity, autocorrelation, and year-fixed effects. For all five regions, we see that lagged changes in systemic risk are highly predictive of aggregate default risk (PD) in the following period. Interestingly, lagged PD is generally not significant. Lagged PD significantly matters for South Asia and Southern Europe & Africa. This suggests that our network measure of systemic risk provides explanatory power over and above the measure of credit risk levels in the economy. In short, interconnectedness matters. We observe the lack of significance for network based metrics for all regions other than South America; this implies that once systemic risk measure is included, the role of network based metrics is diminished. Additional financial variables and asset pricing factors also add explanatory power, resulting in an appreciable increase in R^2 in the predictive regressions for all regions. This suggests a material ability to predict credit quality levels in economies using our new measure of systemic risk.

[Insert Table 12 here]

Table 13 summarizes the contribution to predictability from the various predictor variables. Using lagged values of credit risk, systemic risk, firm-specific network attributes, and U.S. macroeconomic variables, we are able to explain the out-of-sample variation with an accuracy of over 90%. The systemic risk variable adds a lot of explanatory power to the default risk variable. Together, lagged values of default and interconnectedness are able to predict changes in overall default risk quite successfully, offering policy makers early warnings about changes in the credit cycle.

[Insert Table 13 here]

7 Concluding Comments

Systemic risk is interpreted as a risk that has (a) large impact, (b) is widespread, i.e., affects a large number of entities or institutions, and (c) has a ripple effect that endangers the existence of the financial system. In this project, we undertake a large-scale empirical examination of systemic risk among major financial institutions in the emerging markets. We extend the literature on network models by incorporating credit quality information in order to compute a single systemic risk score that summarizes the level of systemic risk across all emerging market financial entities. We provide computations of the dynamics of systemic risk evolution across emerging markets, and study the cross-sectional and time series determinants of systemic risk. We also examine the relationship of systemic risk to macroeconomic and market variables in order to assess if systemic risk and aggregate credit quality in emerging markets may be predicted. Indeed, we find that network measures of risk, including our systemic risk variable, enable prediction of credit risk levels on a quarterly horizon.

Taken together, our findings show that systemic risks for emerging market financial firms, determined jointly by underlying network and credit risks, are quickly transmitted across markets contemporaneously within the same quarter. This implies that regulators have to initiate quick policy actions to manage and stabilize financial markets facing possible systemic risk events. The policy measures should target lowering network and default risks, perhaps through financial easing and short-term liquidity provision measures. Moreover, we find a factor structure among systemic risks across markets, where the first three principal components explain over 90% of the variance, and are tied to US default, funding liquidity, and monetary policy uncertainty factors. Accordingly, regulators could design specific strategies to control systemic risks based on which type of PC dominates that event. Moreover, our network measure of systemic risk can be used by regulators to predict financial sector credit quality changes in emerging markets, thereby offering a macroprudential tool for the management of the credit cycle.

References

- Abbass, P., C. Brownlees, C. Hans, and N. Podlich (2016, April). Credit risk interconnect-
edness: What does the market really know? *Journal of Financial Stability* 29, 1–12.
- Acemoglu, D., A. Ozdaglar, and A. Tahbaz-Salehi (2015). Systemic Risk and Stability in
Financial Networks. *American Economic Review* 105, 564–608.
- Acharya, V., R. Engle, and M. Richardson (2012). Capital Shortfall: A New Approach to
Ranking and Regulating Systemic Risks. *American Economic Review* 102, 59–64.
- Acharya, V., L. Pedersen, T. Philippon, and M. Richardson (2016, January). Measuring
Systemic Risk. *Review of Financial Studies* 30(1), 2–47.
- Acharya, V., P. Schnabl, and G. Suarez (2013). Securitization without Risk Transfer. *Journal
of Financial Economics* 103, 515–536.
- Acharya, V. V., J. A. C. Santos, and T. Yorulmazer (2010, August). Systemic Risk and
Deposit Insurance Premiums. *Economic Policy Review*, 89–100.
- Adrian, T. and N. Boyarchenko (2018, July). Liquidity policies and systemic risk. *Journal
of Financial Intermediation* 35, 45–60.
- Adrian, T. and M. K. Brunnermeier (2016). CoVaR. *American Economic Review* 106(7),
1705–1741.
- Adrian, T. and H. Shin (2010). Liquidity and Leverage. *Journal of Financial Intermedia-
tion* 19, 418–437.
- Ahern, K. R. (2013, December). Network Centrality and the Cross Section of Stock Returns.
SSRN Scholarly Paper ID 2197370, Social Science Research Network, Rochester, NY.
- Albuquerque, R., L. Cabral, and J. Guedes (2019, November). Incentive Pay and Systemic
Risk. *The Review of Financial Studies* 32(11), 4304–4342.
- Allen, F. and E. Carletti (2013). What Is Systemic Risk? *Journal of Money, Credit and
Banking* 45, 121–127.
- Anand, K., P. Gai, S. Kapadia, and S. Brennan (2013). A Network Model of Financial
System Resilience. *Journal of Economic Behavior and Organization* 85, 219–235.
- Anand, K., I. van Lelyveld, d. Banai, S. Friedrich, R. Garratt, G. Hałaj, J. Figue, I. Hansen,
S. M. Jaramillo, H. Lee, J. L. Molina-Borboa, S. Nobili, S. Rajan, D. Salakhova, T. C.
Silva, L. Silvestri, and S. R. S. de Souza (2018, April). The missing links: A global
study on uncovering financial network structures from partial data. *Journal of Financial
Stability* 35, 107–119.
- Anderson, H., M. Paddrik, and J. J. Wang (2019, September). Bank Networks and Systemic
Risk: Evidence from the National Banking Acts. *American Economic Review* 109(9),
3125–3161.
- Aparna Gupta, Y. Lu, and R. Wang (2018). Addressing systemic risk using Contingent
Convertible debt – A Network analysis. *Working Paper, SEC*.

- Avdjiev, S., M. Chui, and H. Shin (2014). Non-financial corporations from emerging market economies and capital flows. *BIS Quarterly Review*.
- Avramidis, P. and F. Pasiouras (2015). Calculating systemic risk capital: A factor model approach. *Journal of Financial Stability* 16, 138–150.
- Benoit, S., J. Colliard, C. Hurlin, and C. Pérignon (2017). Where the Risks Lie: A Survey on Systemic Risk. *Review of Finance* 21(1), 109–152.
- Benoit, S., C. Hurlin, and C. Pérignon (2019). Pitfalls in systemic-risk scoring. *Journal of Financial Intermediation* 38(C), 19–44.
- Berger, A. N., R. A. Roman, and J. Sedunov (2019, January). Did TARP Reduce or Increase Systemic Risk? The Effects of Government Aid on Financial System Stability. *Journal of Financial Intermediation* forthcoming.
- Betz, F., N. Hautsch, T. A. Peltonend, and M. Schienle (2016). Systemic risk spillovers in the European banking and sovereign network. *Journal of Financial Stability* 25, 206–224.
- Bianchi, D., M. Billio, R. Casarin, and M. Guidolin (2019, May). Modeling systemic risk with Markov Switching Graphical SUR models. *Journal of Econometrics* 210(1), 58–74.
- Bianchi, D., M. Billio, R. Casarin, and G. Massimo (2015). Modeling Contagion and Systemic Risk. *SYRTO Working paper*.
- Billio, M., M. Getmansky, D. Gray, A. Lo, R. Merton, and L. Pelizzon (2012). Sovereign, Bank and Insurance Credit Spreads: Connectedness and System Networks. *Working paper, IMF*.
- Billio, M., M. Getmansky, A. W. Lo, and L. Pelizzon (2012). Econometric measures of connectedness and systemic risk in the finance and insurance sectors. *Journal of Financial Economics* 104(3), 535–559.
- Bisias, D., M. Flood, A. Lo, and S. Valavanis (2012). A Survey of Systemic Risk Analytics. *Annual Review of Financial Economics* 4, 255–296.
- Black, L., R. Correa, X. Huang, and H. Zhou (2016). The systemic risk of European banks during the financial and sovereign debt crises. *Journal of Banking and Finance* 63, 107–125.
- Bluhm, M. and J. P. Krahnen (2014). Systemic risk in an interconnected banking system with endogenous asset markets. *Journal of Financial Stability* 13, 75–94.
- Bonacich, P. (1987). Power and Centrality: A Family of Measures. *American Journal of Sociology* 92(5), 1170–1182.
- Bonacich, P. and P. Lloyd (2001, July). Eigenvector-like measures of centrality for asymmetric relations. *Social Networks* 23(3), 191–201.
- Borri, N. (2018). Local currency systemic risk. *Emerging Markets Review* 34(C), 111–123.
- Bostandzic, D. and G. N. F. Weiß (2018, July). Why do some banks contribute more to global systemic risk? *Journal of Financial Intermediation* 35, 17–40.
- Brownlees, C. and R. Engle (2017). SRISK: A Conditional Capital Shortfall Index for Systemic Risk Measurement. *Review of Financial Studies* 30(1), 48–79.

- Brunetti, C., J. H. Harris, S. Mankad, and G. Michailidis (2015). Interconnectedness in the Interbank Market.
- Brunnermeier, M. and L. Pedersen (2009). Market Liquidity and Funding Liquidity. *Review of Financial Studies* 22(6), 2201–2238.
- Buch, C. M., T. Krause, and L. Tonzer (2019). Drivers of systemic risk: Do national and European perspectives differ? *Journal of International Money and Finance* 91(C), 160–176.
- Cai, J., F. Eidam, A. Saunders, and S. Steffen (2018). Syndication, interconnectedness, and systemic risk. *Journal of Financial Stability* 34(C), 105–120.
- Chan-lau, J. A., M. A. Espinosa, K. Giesecke, and J. Solé (2009). Assessing the Systemic Implications of Financial Linkages. In *IMF Working papers*.
- Chan-Lau, J. A., C. Chuang, J.-C. Duan, and W. Sun (2016, May). Banking Network and Systemic Risk via Forward-Looking Partial Default Correlations. *IMF Working Papers*.
- Colliard, J.-E., T. Foucault, and P. Hoffmann (2017). Interbank Trading in a Segmented OTC Market.
- Covitz, D., N. Liang, and G. Suarez (2013). The Evolution of a Financial Crisis: Collapse of the Asset-Backed Commercial Paper Market. *Journal of Finance* 68, 815–848.
- Das, S. R. (2016). Matrix Metrics: Network-Based Systemic Risk Scoring. *Journal of Alternative Investments* 18(4), 33–51.
- Das, S. R., S. R. Kim, and D. N. Ostrov (2019). Dynamic Systemic Risk Networks. *Journal of Financial Data Science* 1, 141–158.
- Das, S. R., K. J. Mitchener, and A. Vossmeier (2018, December). Systemic Risk and the Great Depression. Working Paper 25405, National Bureau of Economic Research.
- Das, S. R. and J. Sisk (2005). Financial Communities. *Journal of Portfolio Management* 31(4), 112–133.
- de Bandt, O. and P. Hartmann (2000, November). Systemic Risk: A Survey. SSRN Scholarly Paper ID 258430, Social Science Research Network, Rochester, NY.
- Demirer, M., F. X. Diebold, L. Liu, and K. Yilmaz (2017). Estimating Global Bank Network Connectedness. *Journal of Applied Econometrics*.
- Dicks, D. L. and P. Fulghieri (2019, June). Uncertainty Aversion and Systemic Risk. *Journal of Political Economy* 127(3), 1118–1155.
- Diebold, F. and K. Yilmaz (2014). On the Network Topology of Variance Decompositions: Measuring the Connectedness of Financial Firms. *Journal of Econometrics* 182, 119–134.
- Dodd, O., M. Kalimipalli, and W. H. Chan (2019, October). Evaluating the Impact of Post-crisis Growth in Emerging Market Corporate Debt. SSRN Scholarly Paper ID 3109160, Social Science Research Network, Rochester, NY.
- Donaldson, J. R. and E. Micheler (2018). Resaleable debt and systemic risk. *Journal of Financial Economics* 127(3), 485–504.

- Duarte, F. and T. Eisenbach (2015, February). Fire-sale spillovers and systemic risk. Staff Reports 645, Federal Reserve Bank of New York.
- Elliott, M., B. Golub, and M. Jackson (2014). Financial Networks and Contagion. *American Economic Review* 104, 3115–3153.
- Ellul, A., C. Jotikasthira, A. V. Kartasheva, C. T. Lundblad, and W. Wagner (2018, January). Insurers As Asset Managers and Systemic Risk. SSRN Scholarly Paper ID 3096147, Social Science Research Network, Rochester, NY.
- Engle, R. (2018). Systemic Risk 10 Years Later. *Annual Review of Financial Economics* 10(1), 125–152.
- Fang, L., B. Sun, H. Li, and H. Yu (2018, June). Systemic risk network of Chinese financial institutions. *Emerging Markets Review* 35, 190–206.
- Freeman, L. (1977). A set of measures of centrality based on betweenness. *Sociometry* 40, 35–41.
- Gabrieli, S. and C.-P. Georg (2014). A network view on interbank market freezes. Discussion Paper 44/2014, Deutsche Bundesbank.
- Gale, D. M. and S. Kariv (2007). Financial Networks. *American Economic Review, Papers and Proceedings*.
- Giglio, S., B. Kelly, and S. Pruitt (2016). Systemic risk and the macro economy: An empirical evaluation. *Journal of Financial Economics* 119(3), 457–471.
- Gobat, J., T. Barnhill, A. Jobst, T. Kisinbay, H. Oura, T. Severo, and L. Schumacher (2011). How to Address the Systemic Part of Liquidity Risk.
- Goodhart, C. (2009, August). Liquidity Management. In *Financial Stability and Macroeconomic Policy Symposium*, Jackson Hole. Federal Reserve Bank of Kansas City.
- Gorton, G. and A. Metrick (2012). Securitized Banking and the Run on Repo. *Journal of Financial Economics* 104, 425–451.
- Hanson, S., A. Kashyap, and J. Stein (2011). A Macroprudential Approach to Financial Regulation. *Journal of Economic Perspectives* 25, 3–28.
- Hautsch, N., J. Schaumburg, and M. Schienle (2015). Financial Network Systemic Risk Contributions. *Review of Finance* 19, 685–738.
- He, Z. and A. Krishnamurthy (2013, April). Intermediary Asset Pricing. *American Economic Review* 103(2), 732–770.
- Härdle, W. K., W. Wang, and L. Yuc (2016). TENET: Tail-Event driven NETWORK risk. *Journal of Econometrics* 192(2), 499–513.
- Huang, X., H. Zhou, and H. Zhu (2012). Systemic Risk Contributions. *Journal of Financial Services Research* 42, 55–83.
- Karolyi, G. A., J. Sedunov, and A. G. Taboada (2018, April). Cross-Border Bank Flows and Systemic Risk. SSRN Scholarly Paper ID 2938544, Social Science Research Network, Rochester, NY.

- Kaserer, C. and C. Klein (2019). Systemic Risk in Financial Markets: How Systemically Important Are Insurers? *Journal of Risk and Insurance* 86(3), 729–759.
- Kitwiwattanachai, C. (2015). Learning Network Structure of Financial Institutions from CDS Data. *University of Connecticut, Working paper*.
- Laeven, L., L. Ratnovski, and H. Tong (2016). Bank size, capital, and systemic risk: Some international evidence. *Journal of Banking and Finance* 69(1), S25–S34.
- Lehar, A. (2005). Measuring Systemic Risk: A Risk Management Approach. *Journal of Banking and Finance* 29, 2577–2603.
- Löffler, G. and P. Raupach (2018, February). Pitfalls in the Use of Systemic Risk Measures. *Journal of Financial and Quantitative Analysis* 53(1), 269–298.
- Li, F. and H. Perez-Saiz (2018). Measuring systemic risk across financial market infrastructures. *Journal of Financial Stability* 34(C), 1–11.
- Li, J. and G. Zinna (2014). On Bank Credit Risk: Systemic or Bank Specific? Evidence for the United States and United Kingdom. *Journal of Financial and Quantitative Analysis* 5/6, 1403–1442.
- Liang, N. (2013). Systemic Risk Monitoring and Financial Stability. *Journal of Money, Credit and Banking* 45, 129–135.
- Liu, S., C. Wu, C.-Y. Yeh, and W. Yoo (2017, July). What Drives Systemic Credit Risk? Evidence from the US State CDS Market. SSRN Scholarly Paper ID 3009455, Social Science Research Network, Rochester, NY.
- Markose, S., S. Giansante, and A. Shaghaghi (2012). ‘Too Interconnected to Fail’ Financial Network of US CDS Market: Topological Fragility and Systemic Risk. *Journal of Economic Behavior and Organization* 83, 627–646.
- Merton, R. C. (1973). Theory of Rational Option Pricing. *Bell Journal of Economics and Management Science* 4, 141–183.
- Nier, E., J. Yang, T. Yorulmazer, and A. Alentorn (2007). Network Models and Financial Stability. *Journal of Economic Dynamics and Control* 31, 2033–2060.
- Nucera, F., B. Schwaab, S. J. Koopman, and A. Lucas (2016). The information in systemic risk rankings. *Journal of Empirical Finance* 38, 461–475.
- Oh, D. H. and A. J. Patton (2018, April). Time-Varying Systemic Risk: Evidence From a Dynamic Copula Model of CDS Spreads. *Journal of Business & Economic Statistics* 36(2), 181–195.
- Pagano, M. S. and J. Sedunov (2016). A comprehensive approach to measuring the relation between systemic risk exposure and sovereign debt. *Journal of Financial Stability* 23, 62–78.
- Perotti, E. and J. Suarez (2009, November). Liquidity risk charges as a macro-prudential tool. *VoX EU* <https://voxeu.org/article/liquidity-risk-charges-macro-prudential-tool>.
- Poledna, S., J. L. Molina-Borboad, S. Martínez-Jaramillod, M. v. d. Leije, and S. Thurner (2015). The multi-layer network nature of systemic risk and its implications for the costs of financial crises. *Journal of Financial Stability* 20, 70–81.

- Roukny, T., S. Battiston, and J. E. Stiglitz (2018, April). Interconnectedness as a source of uncertainty in systemic risk. *Journal of Financial Stability* 35, 93–106.
- Saldías, M. (2013). Systemic risk analysis using forward-looking Distance-to-Default series. *Journal of Financial Stability* 9, 498–517.
- Schwarcz, S. (2008). Systemic Risk. *Georgetown Law Journal* 97, 193–249.
- Sedunov, J. (2016). What is the systemic risk exposure of financial institutions? *Journal of Financial Stability* 24, 71–87.
- Sensoy, A. (2017). Firm size, ownership structure, and systematic liquidity risk: The case of an emerging market. *Journal of Financial Stability* forthcoming.
- Sensoy, A., K. Ozturk, E. Hacıhasanoglu, and B. M. Tabak (2017, December). Not all emerging markets are the same: A classification approach with correlation based networks. *Journal of Financial Stability* 33, 163–186.
- Silva, W., H. Kimura, and A. Sobreiro (2017). An analysis of the literature on systemic financial risk: A survey. *Journal of Financial Stability* 28, 91–114.
- Tasca, P., P. Mavrodiev, and F. Schweitzer (2014). Quantifying the impact of leveraging and diversification on systemic risk. *Journal of Financial Stability* 15, 43–52.
- Varotto, S. and L. Zhao (2018). Systemic risk and bank size. *Journal of International Money and Finance* 82(C), 45–70.
- Wang, G.-J., Z.-Q. Jiang, M. Lin, C. Xie, and H. E. Stanley (2018, June). Interconnectedness and systemic risk of China’s financial institutions. *Emerging Markets Review* 35, 1–18.

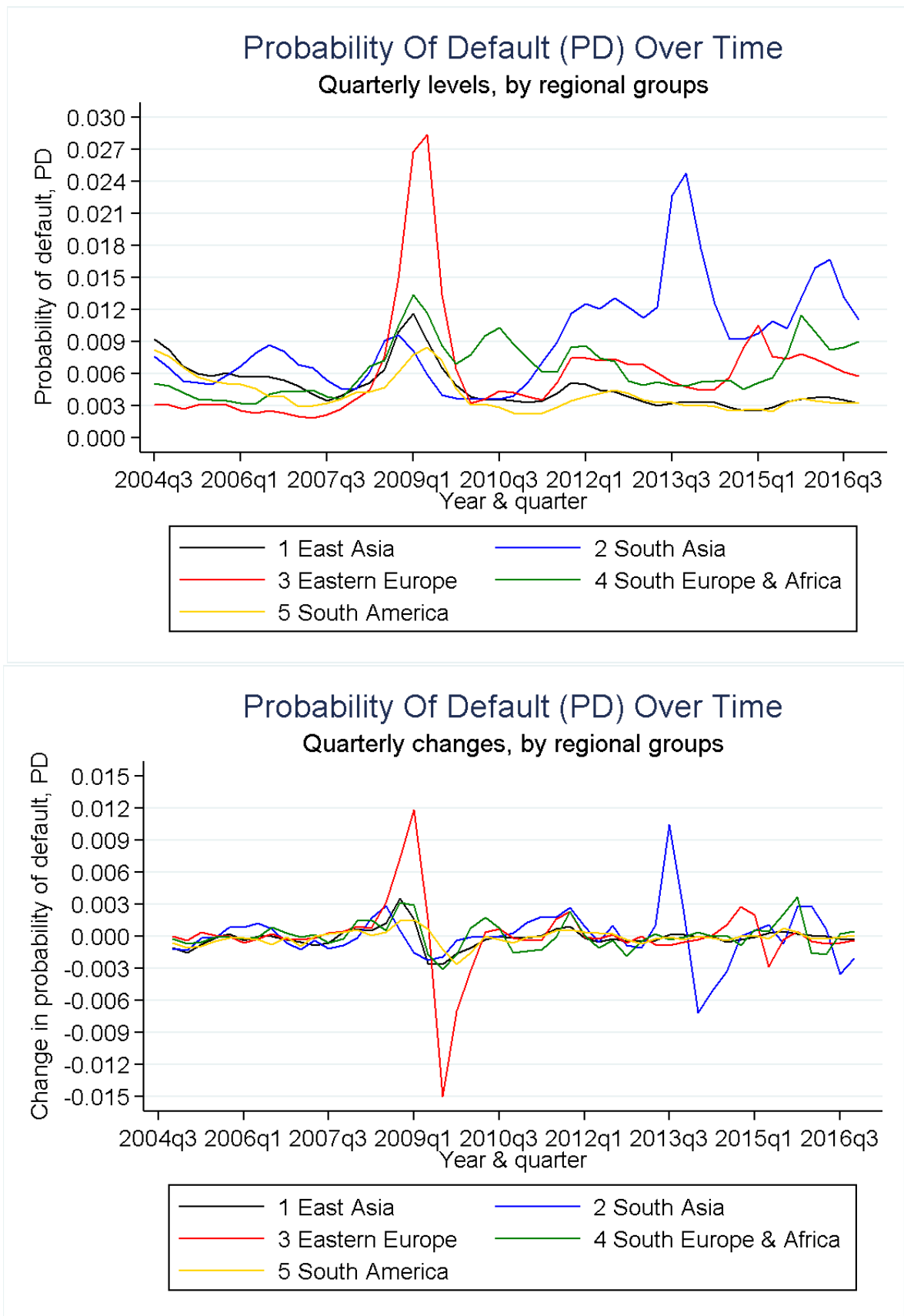


Figure 1: Times series of PD for the entire sample period from 2004/Q3 to 2016/Q4, for each of the five geographic regions. We show the plot in levels and changes.

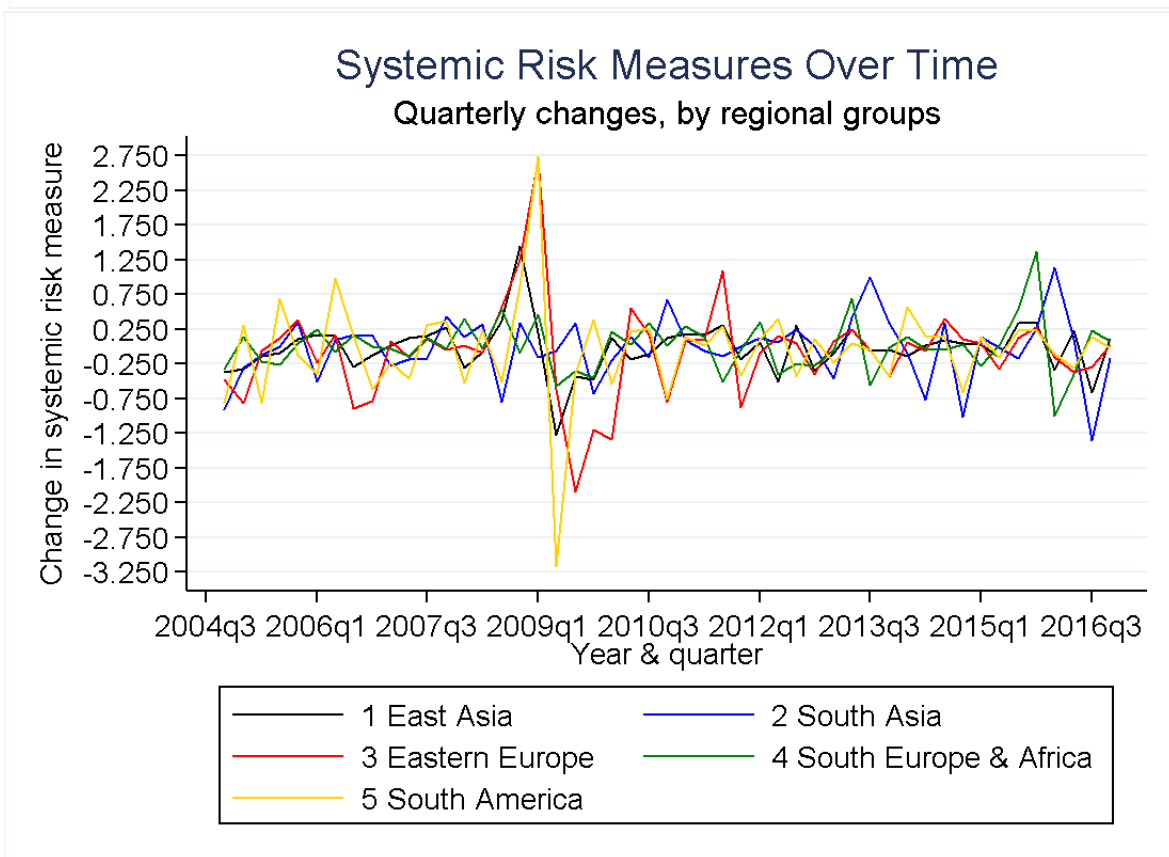
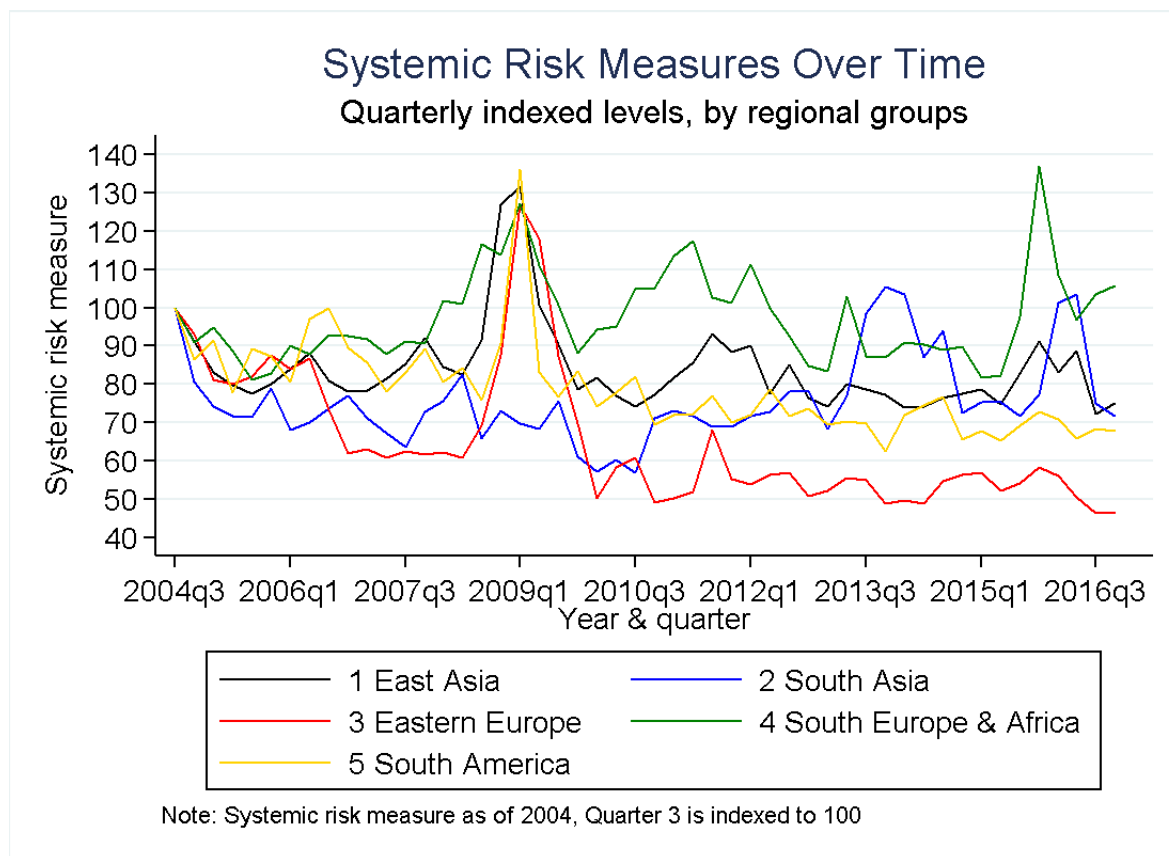
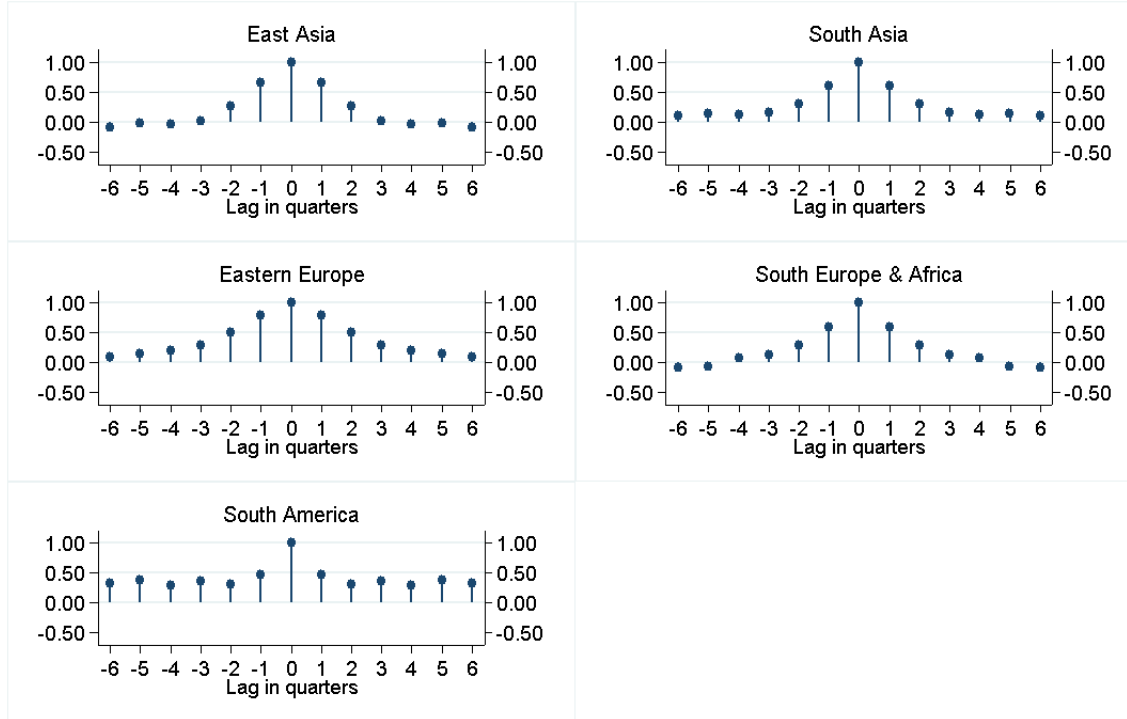


Figure 2: Times series of indexed systemic risk by region. Each series begins at an index value of 100. The data is shown in levels and changes.

Systemic Risk Auto-Correlograms

For quarterly levels, by regional group



Systemic Risk Auto-Correlograms

For quarterly changes, by regional group

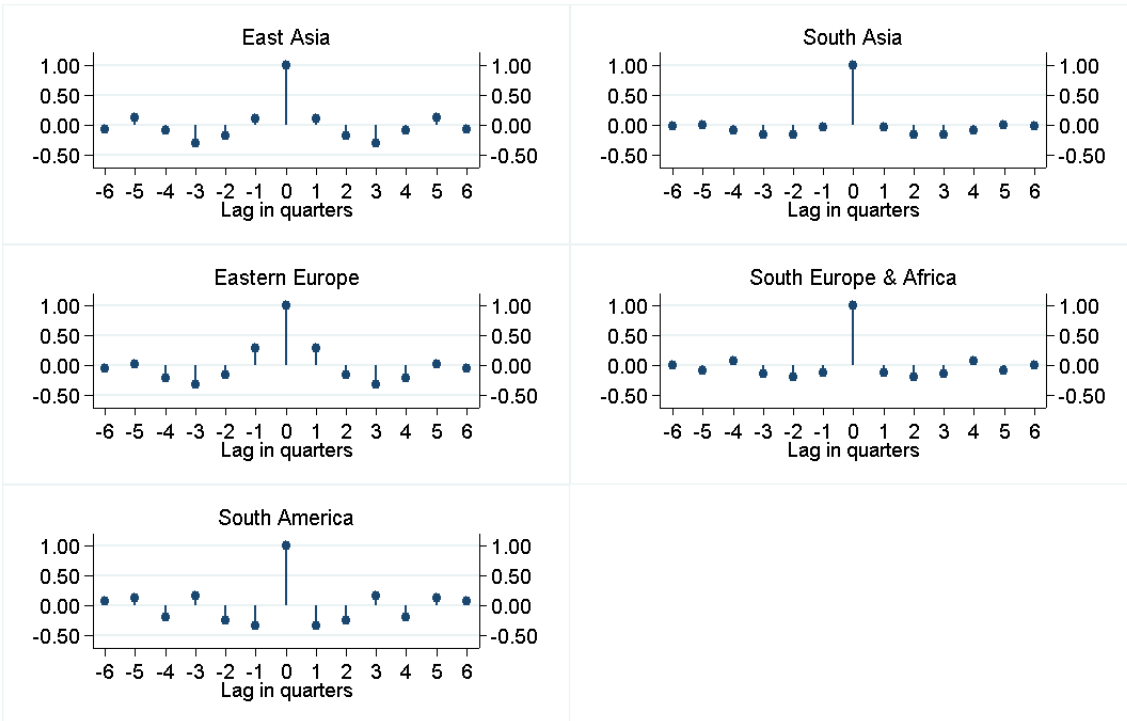
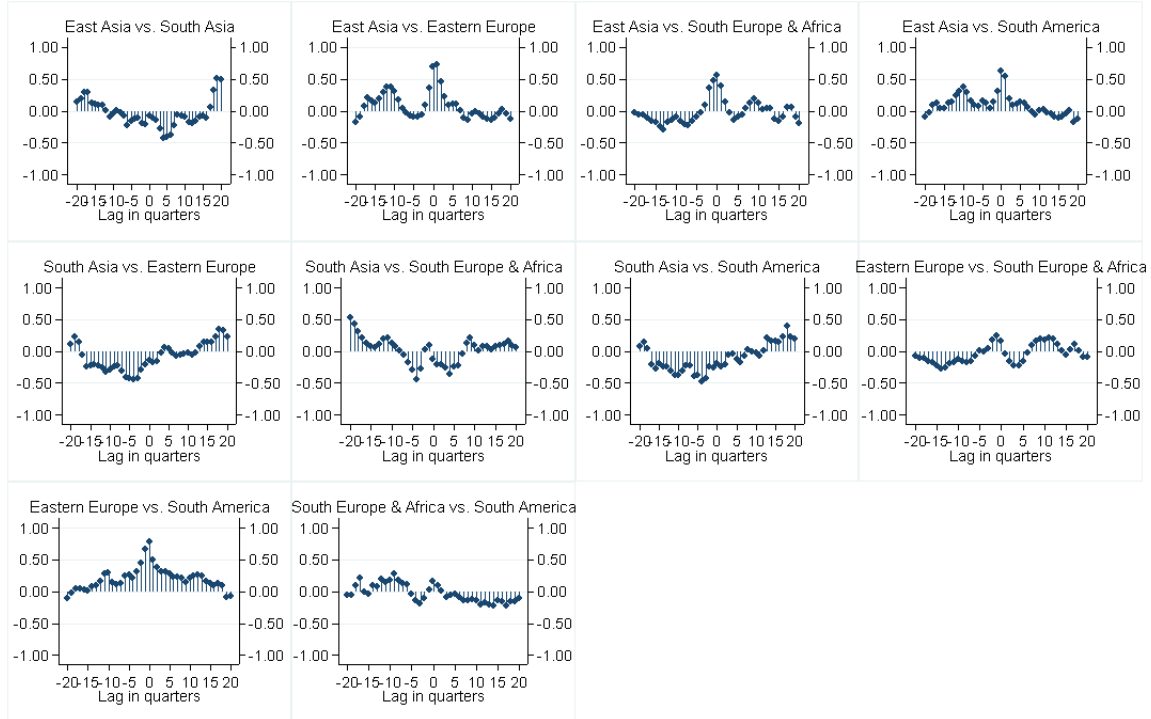


Figure 3: Auto-correlations for the five regions. The data is shown in levels and changes.

Systemic Risk Cross-Correlograms

For quarterly levels, by pairs of regional groups



Systemic Risk Cross-Correlograms

For quarterly changes, by pairs of regional groups

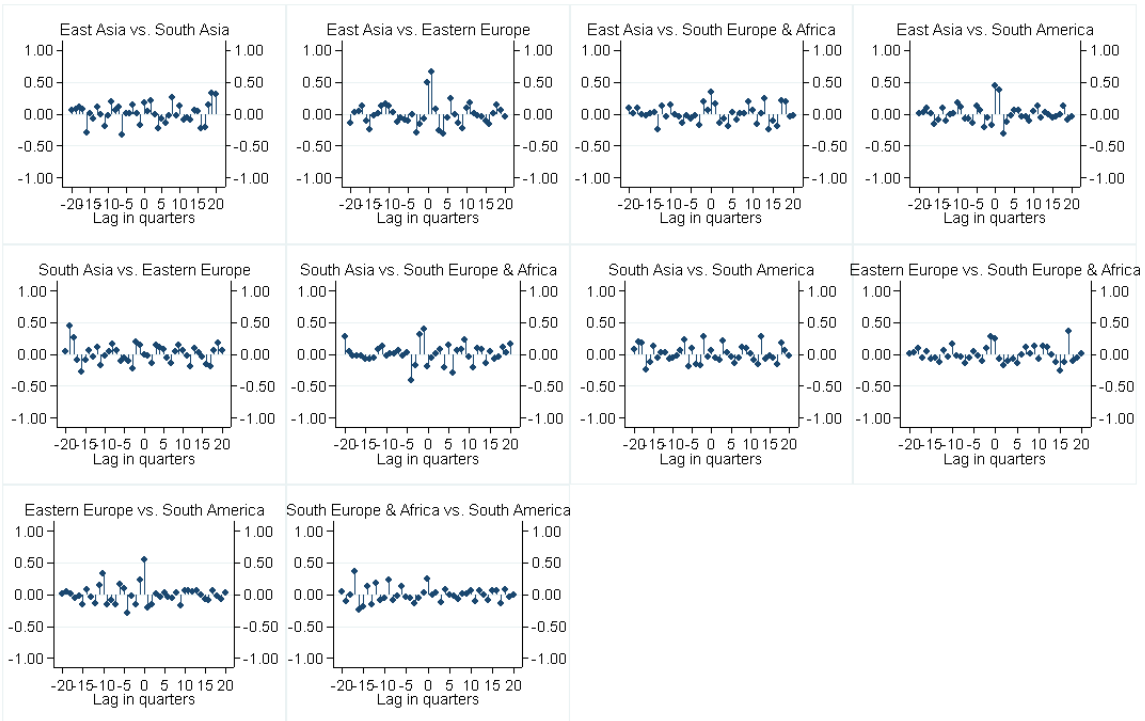


Figure 4: Pairwise cross-correlations between the five regions. There are 10 such plots, one for each pair.

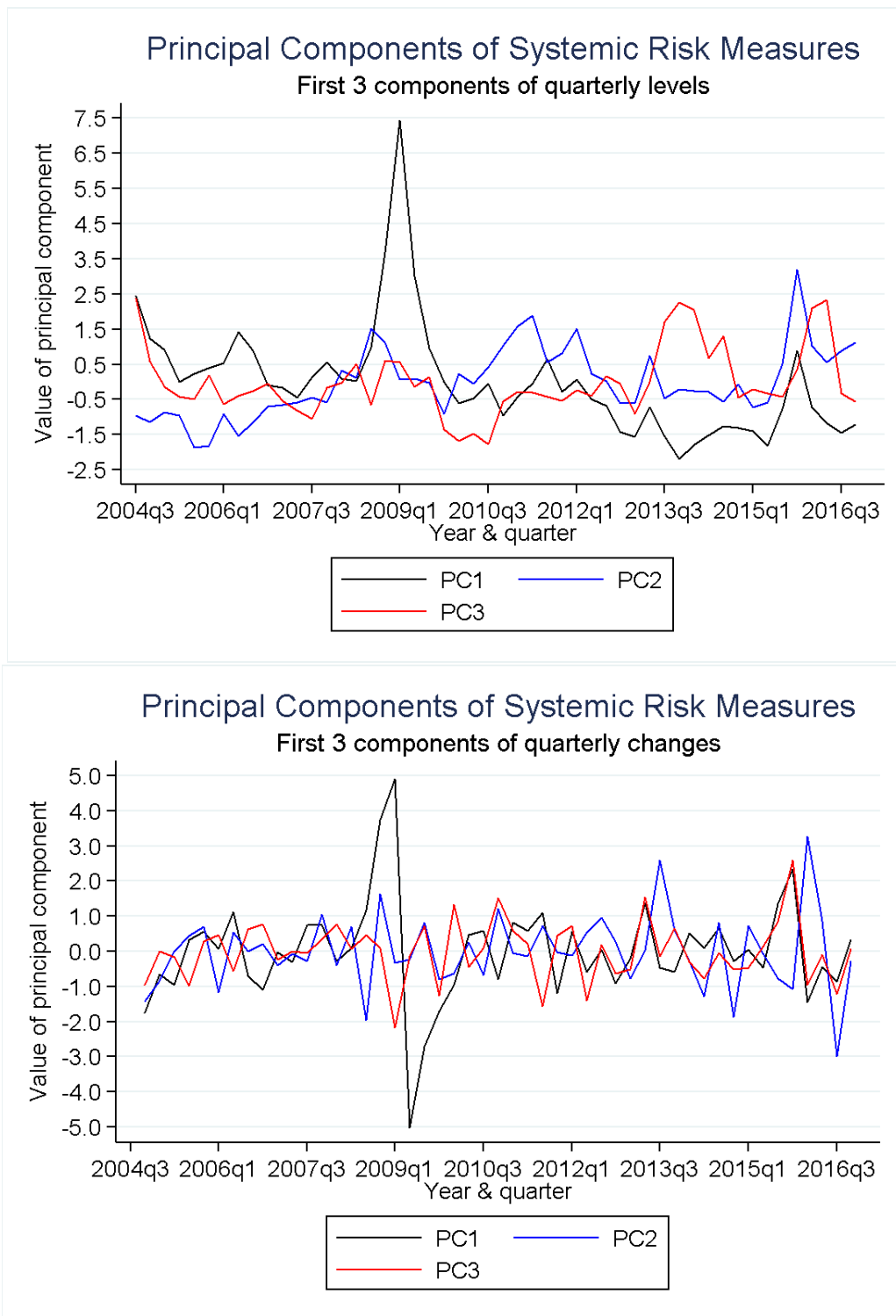


Figure 5: Principal components decomposition of the time series of systemic risk scores for the five regions.

Table 1: Sample count. The table presents the count of industry groups and the number of institutions with valid probability of default (PD) data by country.

Country	Bank	Broker-Dealer	Insurer	Total	Valid PD
Argentina	8	1	0	9	6
Brazil	7	4	6	17	15
Bulgaria	4	5	3	12	9
Chile	8	5	3	16	13
China	26	29	7	62	61
Columbia	8	3	0	11	11
Czech	2	0	0	2	2
Egypt	11	8	2	21	21
Greece	8	3	2	13	12
Hungary	2	3	1	6	5
India	193	191	3	387	356
Indonesia	58	14	14	86	82
Malaysia	15	7	9	31	31
Mexico	9	7	4	20	17
Philippines	28	5	1	34	31
Poland	27	28	5	60	47
Russia	19	2	1	22	16
South Africa	12	12	9	33	32
South Korea	16	24	13	53	52
Taiwan	23	15	12	50	34
Thailand	24	14	17	55	52
Turkey	22	9	6	37	37
Ukraine	9	0	2	11	6
	539	389	120	1,048	948

Table 2: Network measures. The table presents network measures for country regional groups relative to crisis and non-crisis periods. The 23 emerging countries are grouped into five geographical regions. Pre-crisis, crisis, and non-crisis periods encompass years 2004-2006, 2007-2009, and 2010-2016, respectively. Network metrics include number of nodes, mean degree, and mean betweenness centrality between nodes, diameter, fragility, degree concentration measured by HHI, number of distinct clusters, and HHI concentration between clusters. Network connections are based on Granger regressions using p -values of 0.025. All average values reported are computed every quarter.

Geographical region	Period	# of nodes	Mean Degree	Between centrality	Diameter	Fragility	HHI of degree	# of clusters	HHI of clusters
East Asia	Full	217.6	7.38	535.6	12.38	11.53	0.0033	4.18	0.955
	Pre-crisis	151.5	4.53	351.4	15.60	6.97	0.0036	6.30	0.917
	Crisis	176.7	6.71	346.5	12.17	13.14	0.0051	5.25	0.941
	Post-crisis	245.5	8.24	636.9	11.32	12.47	0.0024	2.96	0.975
South Asia	Full	218.7	7.26	630.2	12.30	10.15	0.0022	2.96	0.969
	Pre-crisis	156.5	5.38	414.1	14.60	7.85	0.0036	5.00	0.924
	Crisis	210.7	7.82	558.2	11.67	11.49	0.0023	2.67	0.976
	Post-crisis	236.1	7.49	707.7	11.75	10.39	0.0017	2.36	0.982
Eastern Europe	Full	51.1	1.98	11.4	6.02	3.21	0.0299	11.58	0.410
	Pre-crisis	18.6	0.92	0.3	2.10	1.74	0.0742	9.90	0.087
	Crisis	33.9	1.72	1.5	4.00	3.64	0.0360	11.00	0.367
	Post-crisis	59.0	2.17	15.1	8.29	3.49	0.0131	12.43	0.544
South Europe & Africa	Full	82.9	3.08	42.2	10.28	5.43	0.0097	11.78	0.711
	Pre-crisis	67.9	2.35	17.9	9.30	4.54	0.0133	12.80	0.626
	Crisis	81.5	3.25	48.8	10.25	6.05	0.0103	9.67	0.757
	Post-crisis	87.7	3.22	46.3	10.64	5.48	0.0082	12.32	0.721
South America	Full	39.8	1.54	2.5	4.58	2.87	0.0262	14.06	0.301
	Pre-crisis	29.7	1.41	1.4	3.60	2.64	0.0312	12.40	0.217
	Crisis	32.1	1.57	1.8	3.75	3.01	0.0358	13.58	0.265
	Post-crisis	44.6	1.56	2.9	5.29	2.89	0.0204	14.86	0.346
Overall		174.9	5.94	426.1	9.11	6.65	0.0142	8.91	0.669

Table 3: Risk measures. The table presents the average values of risk measures for country regional groups relative to crisis and non-crisis periods. The 23 emerging countries are grouped into five geographical regions. Pre-crisis, crisis, and non-crisis periods encompass years 2004-2006, 2007-2009, and 2010-2016, respectively. The risk measures include: probability of default, PD and systemic risk score. Network connections are based on Granger regressions using p -values of 0.025. All average values reported are computed every quarter.

Geographical region	Period	# of nodes	Probability of default, PD	Systemic risk score Network-level
East Asia	Full	217.6	0.0045	3.45
	Pre-crisis	151.5	0.0065	3.44
	Crisis	176.7	0.0062	3.83
	Post-crisis	245.5	0.0036	3.28
South Asia	Full	218.7	0.0094	3.59
	Pre-crisis	156.5	0.0068	3.62
	Crisis	210.7	0.0062	3.33
	Post-crisis	236.1	0.0112	3.69
Eastern Europe	Full	51.1	0.0066	4.45
	Pre-crisis	18.6	0.0028	5.65
	Crisis	33.9	0.0098	5.28
	Post-crisis	59.0	0.0062	3.66
South Europe & Africa	Full	82.9	0.0066	3.41
	Pre-crisis	67.9	0.0040	3.15
	Crisis	81.5	0.0073	3.55
	Post-crisis	87.7	0.0071	3.44
South America	Full	39.8	0.0040	4.74
	Pre-crisis	29.7	0.0056	5.40
	Crisis	32.1	0.0051	5.24
	Post-crisis	44.6	0.0032	4.29
Overall		174.9	0.0067	3.93

Table 4: Systemic risk decomposition. The table presents the systemic risk decomposition by contributing institutions for country regional groups relative to crisis and non-crisis periods. The 23 emerging countries are grouped into five geographical regions. Pre-crisis, crisis, and non-crisis periods encompass years 2004-2006, 2007-2009, and 2010-2016, respectively. Panel A reports the average contribution, as a percentage of total regional systemic risk, by the top 10 contributing institutions of each region. Panel B reports the average number of contributing institutions that account for top 50 percentile (that is, contribute at least 50 percent) of total regional systemic risk. Network connections are based on Granger regressions using p -values of 0.025. All values reported denote averages computed every quarter.

Geographical region	Period			
	Full	Pre-crisis	Crisis	Post-crisis
Panel A: Percentage contribution of top 10 contributors				
East Asia	16.18	18.54	20.69	13.41
South Asia	20.09	21.50	17.94	20.51
Eastern Europe	47.11	72.89	52.83	35.45
South Europe & Africa	31.65	33.52	33.46	30.21
South America	46.72	55.79	51.48	41.45
Panel B: Number of contributors in top 50 percentile				
East Asia	54.9	42.1	41.6	65.1
South Asia	41.7	36.3	46.2	41.7
Eastern Europe	12.1	5.8	8.8	15.8
South Europe & Africa	20.2	18.3	18.7	21.6
South America	10.8	8.0	9.1	12.5

Table 5: Time-series regressions of quarterly changes in regional systemic risk. Panels A through E report the results for individual geographical regions. The dependent variable is the difference in values of network level systemic risk score between quarters t and $t - 1$. Primary explanatory variables include: quarterly changes in credit risk (mean probability of default, PD); and quarterly changes in network attributes (mean degree across all nodes, degree concentration measured by HHI, mean centrality between nodes, diameter, fragility, number of distinct clusters, and HHI concentration within clusters). Additional control variables include: quarterly changes in median firm-specific attributes, set 1 (book value of assets, loans-to-assets and loans-to-deposits ratios of banks, debt-to-assets ratios, debt-to-capital ratio, returns on assets, and market-to-book value of equity); quarterly changes in median firm-specific attributes, set 2 (market value of equity, loans-to-assets and loans-to-deposits ratios of banks, debt-to-equity ratios, debt-to-capital ratio, returns on equity, and market-to-book value of equity); and quarterly changes in U.S. macroeconomic variables (default factor, level and slope of term structure factor, TED spread, and VIX). Network connections are based on Granger regressions using p-values of 0.025. Mean PD, degree and centrality across firms, and market-wide median firm-specific attributes are computed every quarter. All regressions include adjustments for heteroscedasticity, autocorrelation in residuals and year fixed-effects. t -statistics are reported in parentheses. Coefficients marked by *, **, and *** denote $p < 0.05$, $p < 0.01$, and $p < 0.001$, respectively.

Panel A: East Asia								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Constant	0.0603 (0.33)	-0.2800*** (-6.11)	-0.3674*** (-6.11)	-0.0926 (-1.44)	-0.0979 (-1.43)	-0.1086 (-1.43)	-0.1512 (-1.09)	-0.2699* (-2.26)
Mean PD	317.6259* (2.40)			182.9183*** (5.81)	132.4531*** (4.03)	138.2129** (3.56)	173.2766 (1.89)	104.5822 (1.62)
Mean Degree		0.1871*** (6.86)		0.0640 (0.70)	0.0917 (1.19)	0.0914 (1.14)	0.0325 (0.32)	-0.0615 (-0.68)
Mean Bet. Centrality		-0.0005** (-2.89)		-0.0003 (-1.43)	-0.0001 (-0.29)	-0.0001 (-0.28)	-0.0003 (-1.65)	-0.0004 (-2.03)
Degree HHI			-115.0829** (-3.30)	-76.3943 (-1.25)	-96.4063 (-1.89)	-93.7020 (-1.67)	-182.4880* (-2.46)	-253.4163** (-3.33)
Diameter			-0.0121 (-1.45)	-0.0098 (-0.99)	-0.0077 (-0.81)	-0.0072 (-0.78)	-0.0035 (-0.38)	-0.0072 (-1.01)
Fragility			0.0947*** (6.70)	0.0541 (1.25)	0.0504 (1.42)	0.0527 (1.41)	0.0878 (1.71)	0.1372** (2.97)
Num. Clusters			0.0231 (0.49)	-0.0004 (-0.01)	0.0334 (1.49)	0.0293 (1.17)	0.0789* (2.90)	0.0689** (3.07)
Cluster HHI			2.7984 (0.69)	0.5930 (0.22)	3.5628 (1.71)	3.0250 (1.26)	7.0655** (3.29)	7.0324** (2.98)
Control variables included:								
Firm attributes, set 1	No	No	No	No	Yes	No	Yes	No
Firm attributes, set 2	No	No	No	No	No	Yes	No	Yes
U.S. macro variables	No	No	No	No	No	No	Yes	Yes
Observations	49	49	49	49	49	49	49	49
R^2	0.617	0.906	0.908	0.963	0.985	0.984	0.994	0.994
Adjusted R^2	0.475	0.867	0.858	0.936	0.967	0.964	0.982	0.982

Panel B: South Asia								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Constant	-0.5223*** (-3.78)	-0.3801*** (-4.79)	-0.3035 (-1.52)	-0.0940 (-0.60)	-0.0542 (-0.29)	-0.0928 (-0.52)	-0.0276 (-0.21)	-0.0468 (-0.36)
Mean PD	66.2983*** (3.88)			114.6030*** (4.57)	0.1945 (0.01)	83.6031*** (3.88)	18.8908 (0.67)	91.1587*** (5.45)
Mean Degree		0.0796** (3.39)		0.2796* (2.09)	-0.1034 (-0.80)	0.0900 (0.74)	-0.1453 (-1.41)	0.0151 (0.15)
Mean Bet. Centrality		-0.0019** (-3.04)		-0.0006 (-1.41)	-0.0012* (-2.40)	-0.0005 (-0.88)	-0.0010* (-2.38)	-0.0003 (-0.72)
Degree HHI			101.5230 (1.70)	109.6812 (1.42)	119.2592 (1.12)	101.6930 (0.98)	137.1667 (2.08)	114.0377 (1.77)
Diameter			0.0005 (0.05)	-0.0010 (-0.13)	0.0116 (0.61)	-0.0059 (-0.32)	-0.0002 (-0.02)	-0.0164 (-1.05)
Fragility			0.0625*** (4.87)	-0.0859 (-1.12)	0.1271 (1.58)	0.0299 (0.37)	0.1413* (2.37)	0.0680 (1.17)
Num. Clusters			0.0524 (0.58)	-0.0989 (-1.21)	-0.0352 (-0.44)	-0.0827 (-0.86)	-0.0231 (-0.29)	-0.0849 (-1.39)
Cluster HHI			-0.0460 (-0.01)	-7.3513 (-1.15)	-5.7838 (-0.93)	-7.5010 (-0.98)	-2.5543 (-0.43)	-6.4160 (-1.22)
Control variables included:								
Firm attributes, set 1	No	No	No	No	Yes	No	Yes	No
Firm attributes, set 2	No	No	No	No	No	Yes	No	Yes
U.S. macro variables	No	No	No	No	No	No	Yes	Yes
Observations	49	49	49	49	49	49	49	49
R^2	0.566	0.693	0.719	0.889	0.973	0.960	0.990	0.989
Adjusted R^2	0.404	0.567	0.565	0.810	0.939	0.909	0.970	0.968

Panel C: Eastern Europe								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Constant	-0.6515*** (-8.46)	-0.4471*** (-4.02)	0.3617 (0.82)	-0.0336 (-0.07)	-0.0572 (-0.09)	-0.2103 (-0.42)	-0.5516 (-0.79)	0.0520 (0.07)
Mean PD	147.3393*** (5.29)			140.8717*** (6.07)	117.6743*** (4.45)	116.0490*** (4.88)	72.9176* (2.43)	85.1026 (1.94)
Mean Degree		0.4181** (3.55)		0.4528 (1.69)	0.6306 (1.59)	0.7411 (1.83)	1.9052*** (4.61)	1.0048 (1.73)
Mean Bet. Centrality		-0.0004 (-0.11)		-0.0028 (-0.55)	-0.0027 (-0.36)	-0.0055 (-0.73)	-0.0373* (-2.75)	-0.0119 (-0.95)
Degree HHI			-9.6052 (-1.80)	-2.9613 (-0.45)	0.2882 (0.03)	3.4239 (0.50)	7.5327 (0.69)	-1.2824 (-0.11)
Diameter			0.0168 (0.96)	0.0044 (0.18)	0.0054 (0.20)	0.0216 (0.80)	0.1112* (2.69)	0.0617 (1.97)
Fragility			0.3278** (3.64)	-0.0080 (-0.06)	-0.0586 (-0.31)	-0.1811 (-1.11)	-0.4513* (-2.46)	-0.1259 (-0.53)
Num. Clusters			-0.0432 (-1.70)	-0.0217 (-1.03)	-0.0334 (-1.23)	-0.0294 (-1.15)	-0.0305 (-0.76)	-0.0305 (-0.85)
Cluster HHI			-1.1542 (-1.21)	-0.5420 (-1.17)	-1.0999 (-1.46)	-0.8867 (-1.40)	-2.4493** (-3.37)	-1.7417 (-1.83)
Control variables included:								
Firm attributes, set 1	No	No	No	No	Yes	No	Yes	No
Firm attributes, set 2	No	No	No	No	No	Yes	No	Yes
U.S. macro variables	No	No	No	No	No	No	Yes	Yes
Observations	49	49	47	47	47	47	47	47
R^2	0.823	0.709	0.731	0.944	0.969	0.979	0.990	0.986
Adjusted R^2	0.757	0.590	0.573	0.901	0.925	0.948	0.968	0.955

Panel D: South Europe & Africa								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Constant	-0.0199 (-0.14)	-0.1154 (-1.15)	-0.1331 (-1.04)	-0.0034 (-0.07)	-0.1135 (-1.41)	-0.1330 (-1.98)	-0.1994* (-2.12)	-0.1743* (-2.21)
Mean PD	158.8254* (2.43)			127.6448*** (9.35)	127.7732*** (10.07)	129.9267*** (9.18)	121.6046*** (4.79)	110.5545** (3.59)
Mean Degree		0.3694*** (9.00)		0.5661*** (5.78)	0.4842** (3.57)	0.4946*** (4.84)	0.4955** (3.65)	0.5348*** (4.38)
Mean Bet. Centrality		-0.0001 (-0.20)		-0.0003 (-0.60)	-0.0005 (-0.64)	-0.0004 (-0.54)	-0.0007 (-0.57)	-0.0008 (-0.64)
Degree HHI			-41.4282*** (-4.83)	26.2980* (2.72)	24.5068 (1.80)	27.6075* (2.68)	22.5185 (1.61)	30.7914* (2.30)
Diameter			0.0054 (0.49)	0.0060 (0.92)	0.0123 (1.60)	0.0145 (1.97)	0.0126 (1.13)	0.0189 (1.69)
Fragility			0.2142*** (7.38)	-0.1136* (-2.15)	-0.0699 (-0.94)	-0.0845 (-1.50)	-0.0716 (-0.94)	-0.0919 (-1.29)
Num. Clusters			0.0149 (0.74)	0.0186 (1.74)	0.0091 (0.64)	0.0130 (0.97)	0.0206 (1.02)	0.0210 (1.19)
Cluster HHI			0.0831 (0.10)	0.6163 (1.39)	0.2174 (0.32)	0.4346 (0.70)	0.6823 (0.76)	0.7061 (0.90)
Control variables included:								
Firm attributes, set 1	No	No	No	No	Yes	No	Yes	No
Firm attributes, set 2	No	No	No	No	No	Yes	No	Yes
U.S. macro variables	No	No	No	No	No	No	Yes	Yes
Observations	49	49	49	49	49	49	49	49
R^2	0.502	0.828	0.824	0.973	0.982	0.984	0.986	0.987
Adjusted R^2	0.317	0.757	0.728	0.954	0.959	0.963	0.957	0.961

Panel E: South America								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Constant	-0.4839 (-1.06)	-0.2571 (-1.65)	-0.2551* (-2.19)	0.0499 (0.98)	0.1537* (2.39)	0.1708* (2.35)	0.0827 (1.24)	0.1183 (1.70)
Mean PD	-176.3660 (-0.48)			288.8499*** (8.30)	300.1950*** (7.11)	309.0190*** (8.43)	360.2092*** (6.76)	349.2268*** (7.17)
Mean Degree		0.7176*** (13.29)		0.3780 (1.97)	0.2775 (1.34)	0.3414 (1.56)	0.3990 (1.81)	0.3096 (1.50)
Mean Bet. Centrality		-0.0115 (-1.12)		-0.0231* (-2.75)	-0.0019 (-0.22)	-0.0062 (-0.73)	-0.0051 (-0.53)	-0.0073 (-0.81)
Degree HHI			-16.9225** (-3.30)	-8.4369 (-1.51)	-10.7583 (-1.50)	-9.1713 (-1.26)	-8.9602 (-1.20)	-10.1882 (-1.62)
Diameter			-0.0233 (-0.98)	0.0329* (2.11)	0.0142 (0.75)	0.0221 (1.23)	0.0253 (1.30)	0.0403* (2.27)
Fragility			0.4499*** (6.27)	0.2601* (2.13)	0.2968* (2.39)	0.2620 (1.98)	0.2523 (1.94)	0.2768* (2.33)
Num. Clusters			-0.0077 (-0.50)	-0.0219** (-2.96)	-0.0266* (-2.50)	-0.0282* (-2.42)	-0.0162 (-1.66)	-0.0194 (-1.81)
Cluster HHI			-0.2181 (-0.78)	-0.5769*** (-4.19)	-0.6379** (-3.30)	-0.6909** (-3.15)	-0.4887* (-2.88)	-0.5220* (-2.57)
Control variables included:								
Firm attributes, set 1	No	No	No	No	Yes	No	Yes	No
Firm attributes, set 2	No	No	No	No	No	Yes	No	Yes
U.S. macro variables	No	No	No	No	No	No	Yes	Yes
Observations	49	49	49	49	49	49	49	49
R^2	0.265	0.948	0.937	0.984	0.993	0.993	0.995	0.995
Adjusted R^2	-0.008	0.926	0.902	0.973	0.985	0.985	0.986	0.986

Table 6: Panel regressions of quarterly systemic risk contributions of firms. Panel A reports the results for all geographical regions considered jointly, and Panels B through F present the results for individual geographical regions. The dependent variable is each firm's contribution to network level systemic risk score. Primary explanatory variables include: credit risk (probability of default, PD); and network attributes (degree across all nodes, degree concentration measured by HHI, centrality between nodes, diameter, fragility, number of distinct clusters, and HHI concentration within clusters). Additional control variables include: firm-specific attributes, set 1 (book value of assets, loans-to-assets and loans-to-deposits ratios of banks, debt-to-assets ratios, debt-to-capital ratio, returns on assets, and market-to-book value of equity); firm-specific attributes, set 2 (market value of equity, loans-to-assets and loans-to-deposits ratios of banks, debt-to-equity ratios, debt-to-capital ratio, returns on equity, and market-to-book value of equity); and U.S. macroeconomic variables (default factor, level and slope of term structure factor, TED spread, and VIX). Network connections are based on Granger regressions using p-values of 0.025. All regressions include adjustments for heteroscedasticity and year fixed-effects. Panel A regressions also include adjustments for region-specific cluster-effects. t -statistics are reported in parentheses. Coefficients marked by *, **, and *** denote $p < 0.05$, $p < 0.01$, and $p < 0.001$, respectively.

Panel A: All Regions								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Constant	1.2100 (2.46)	1.3349 (2.39)	4.1943** (5.99)	3.9919** (5.72)	2.9974* (4.07)	3.4410* (4.21)	2.7916* (4.43)	3.1815** (4.61)
PD	23.8580* (2.87)			24.2650* (3.10)	16.3571 (2.63)	17.8211* (2.83)	16.1400 (2.68)	17.5494* (2.87)
Degree		0.0255 (1.55)		0.0760* (2.98)	0.0781 (2.69)	0.0786 (2.57)	0.0783 (2.71)	0.0789 (2.59)
Bet. Centrality		-0.0002 (-1.33)		-0.0002 (-1.26)	-0.0002 (-1.15)	-0.0002 (-1.10)	-0.0002 (-1.15)	-0.0002 (-1.10)
Degree HHI			23.7614** (5.55)	24.0806** (5.09)	23.1830** (4.86)	24.8355* (4.48)	23.1151** (4.97)	24.7272* (4.55)
Diameter			-0.0132 (-0.95)	-0.0097 (-0.70)	-0.0241 (-1.39)	-0.0237 (-1.36)	-0.0238 (-1.44)	-0.0231 (-1.39)
Fragility			-0.0275 (-1.67)	-0.0556 (-2.05)	-0.0782 (-2.04)	-0.0785 (-1.99)	-0.0814 (-2.20)	-0.0829 (-2.17)
Num. Clusters			-0.0594* (-3.25)	-0.0548* (-2.85)	-0.0536* (-2.87)	-0.0508 (-2.54)	-0.0529* (-2.79)	-0.0499 (-2.45)
Cluster HHI			-2.9421** (-5.73)	-3.1186** (-6.59)	-2.7347* (-3.87)	-2.6743* (-3.27)	-2.6955* (-3.78)	-2.6246* (-3.19)
Control variables included:								
Firm attributes, set 1	No	No	No	No	Yes	No	Yes	No
Firm attributes, set 2	No	No	No	No	No	Yes	No	Yes
U.S. macro variables	No	No	No	No	No	No	Yes	Yes
Observations	29158	29158	29145	29145	19946	17866	19946	17866
R^2	0.092	0.039	0.474	0.644	0.681	0.679	0.682	0.681
Adjusted R^2	0.091	0.039	0.474	0.644	0.681	0.679	0.682	0.680

Panel B: East Asia								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Constant	0.4804*** (16.85)	0.4624*** (15.87)	1.6920 (1.76)	1.0066* (2.10)	0.7947 (1.92)	1.1105* (2.56)	0.7217 (1.67)	1.0419* (2.32)
PD	22.5379*** (19.55)			21.1721*** (24.72)	19.6228*** (25.62)	20.9975*** (27.12)	19.6635*** (25.60)	21.0402*** (27.09)
Degree		0.0432*** (32.33)		0.0455*** (38.34)	0.0454*** (38.00)	0.0452*** (37.21)	0.0454*** (37.95)	0.0452*** (37.15)
Bet. Centrality		0.0000** (2.59)		-0.0000 (-0.34)	0.0000 (0.36)	0.0000 (0.65)	0.0000 (0.35)	0.0000 (0.64)
Degree HHI			4.2298 (0.51)	20.6064*** (4.57)	22.6657*** (5.46)	23.2560*** (5.35)	23.3707*** (5.34)	23.9715*** (5.16)
Diameter			0.0023 (0.84)	0.0032 (1.94)	0.0037* (2.54)	0.0037* (2.47)	0.0038* (2.55)	0.0038* (2.47)
Fragility			-0.0004 (-0.16)	-0.0231*** (-17.92)	-0.0227*** (-18.17)	-0.0224*** (-17.29)	-0.0228*** (-17.80)	-0.0224*** (-17.05)
Num. Clusters			-0.0134 (-1.25)	-0.0066 (-1.24)	-0.0110* (-2.38)	-0.0114* (-2.37)	-0.0098* (-1.99)	-0.0103* (-2.03)
Cluster HHI			-1.0643 (-1.08)	-0.6997 (-1.41)	-1.1191** (-2.61)	-1.1875** (-2.64)	-1.1089* (-2.49)	-1.1848* (-2.56)
Control variables included:								
Firm attributes, set 1	No	No	No	No	Yes	No	Yes	No
Firm attributes, set 2	No	No	No	No	No	Yes	No	Yes
U.S. macro variables	No	No	No	No	No	No	Yes	Yes
Observations	10335	10335	10335	10335	8914	8557	8914	8557
R^2	0.215	0.614	0.078	0.750	0.825	0.824	0.825	0.824
Adjusted R^2	0.214	0.614	0.077	0.750	0.825	0.823	0.825	0.823

Panel C: South Asia								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Constant	0.7508*** (16.99)	0.6378*** (18.19)	1.1737 (0.95)	2.3087*** (3.91)	3.6940** (3.01)	3.8719** (2.80)	4.9622*** (3.85)	5.4284*** (3.75)
PD	20.6470*** (49.56)			18.7515*** (57.58)	16.4707*** (31.74)	17.5219*** (34.31)	16.5935*** (32.21)	17.6579*** (34.85)
Degree		0.0595*** (31.47)		0.0554*** (35.87)	0.0605*** (27.95)	0.0662*** (29.37)	0.0604*** (27.88)	0.0661*** (29.27)
Bet. Centrality		-0.0001*** (-9.96)		-0.0000*** (-5.48)	-0.0000** (-3.26)	-0.0000*** (-3.43)	-0.0000** (-3.26)	-0.0000*** (-3.41)
Degree HHI			33.3122** (2.84)	38.7064*** (5.79)	69.2215*** (5.26)	90.0941*** (5.74)	69.2474*** (4.07)	91.9216*** (4.52)
Diameter			-0.0011 (-0.39)	0.0029 (1.89)	-0.0044 (-1.32)	-0.0020 (-0.54)	-0.0056 (-1.65)	-0.0038 (-1.00)
Fragility			-0.0060** (-2.94)	-0.0317*** (-22.59)	-0.0435*** (-15.73)	-0.0516*** (-15.11)	-0.0450*** (-15.07)	-0.0531*** (-14.85)
Num. Clusters			-0.0007 (-0.05)	-0.0377*** (-5.09)	-0.0576*** (-3.84)	-0.0620*** (-3.61)	-0.0701*** (-4.42)	-0.0759*** (-4.26)
Cluster HHI			-0.4727 (-0.38)	-1.8120** (-3.03)	-3.4284** (-2.75)	-3.2852* (-2.34)	-4.5975*** (-3.53)	-4.8175** (-3.29)
Control variables included:								
Firm attributes, set 1	No	No	No	No	Yes	No	Yes	No
Firm attributes, set 2	No	No	No	No	No	Yes	No	Yes
U.S. macro variables	No	No	No	No	No	No	Yes	Yes
Observations	10609	10609	10609	10609	4329	3375	4329	3375
R^2	0.504	0.378	0.040	0.801	0.847	0.850	0.848	0.852
Adjusted R^2	0.504	0.378	0.039	0.800	0.846	0.849	0.846	0.850

Panel D: Eastern Europe								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Constant	7.5096*** (15.88)	7.3094*** (17.14)	7.9651*** (12.90)	7.3281*** (14.17)	4.1742*** (8.22)	5.7390*** (10.95)	4.0923*** (7.21)	5.5402*** (9.19)
PD	59.0569*** (7.97)			66.6114*** (10.15)	58.4729*** (10.75)	61.4278*** (7.10)	63.9747*** (10.62)	66.6223*** (6.88)
Degree		0.4980*** (21.48)		0.5450*** (28.70)	0.5212*** (27.56)	0.5056*** (25.53)	0.5235*** (26.96)	0.5086*** (25.06)
Bet. Centrality		-0.0047*** (-4.86)		-0.0039*** (-5.04)	-0.0039*** (-5.68)	-0.0029*** (-4.40)	-0.0043*** (-5.99)	-0.0033*** (-4.76)
Degree HHI			6.0075 (1.45)	7.6253* (2.27)	6.5816* (2.02)	7.0739* (2.13)	6.8184* (2.02)	7.4960* (2.19)
Diameter			-0.0088 (-0.61)	-0.0127 (-1.24)	-0.0104 (-1.44)	-0.0153* (-2.03)	-0.0094 (-1.25)	-0.0136 (-1.73)
Fragility			-0.0495 (-0.88)	-0.4036*** (-10.33)	-0.3960*** (-12.10)	-0.4001*** (-11.14)	-0.3018*** (-8.64)	-0.3170*** (-8.56)
Num. Clusters			-0.0726** (-3.15)	-0.0287 (-1.69)	-0.0255 (-1.82)	-0.0233 (-1.53)	-0.0376* (-2.36)	-0.0354* (-2.06)
Cluster HHI			-1.8006** (-2.60)	-0.8765 (-1.80)	-0.7086 (-1.72)	-0.7706 (-1.66)	-1.3717** (-2.81)	-1.4201** (-2.65)
Control variables included:								
Firm attributes, set 1	No	No	No	No	Yes	No	Yes	No
Firm attributes, set 2	No	No	No	No	No	Yes	No	Yes
U.S. macro variables	No	No	No	No	No	No	Yes	Yes
Observations	2186	2186	2173	2173	1733	1443	1733	1443
R^2	0.486	0.644	0.413	0.726	0.865	0.873	0.869	0.876
Adjusted R^2	0.483	0.642	0.408	0.724	0.863	0.871	0.866	0.873

Panel E: South Europe & Africa								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Constant	1.3226*** (18.36)	1.1640*** (17.08)	1.8976** (3.15)	0.8024** (2.81)	-0.5928* (-2.46)	0.2025 (0.78)	-0.3190 (-1.20)	0.4393 (1.55)
PD	60.4727*** (26.90)			60.2266*** (31.15)	58.3385*** (24.31)	65.3833*** (28.73)	59.0894*** (24.83)	66.1804*** (28.94)
Degree		0.2163*** (27.87)		0.2247*** (46.01)	0.2170*** (46.95)	0.2219*** (46.09)	0.2168*** (47.96)	0.2217*** (47.16)
Bet. Centrality		-0.0003 (-1.65)		-0.0004*** (-3.47)	-0.0001 (-1.04)	-0.0001 (-1.41)	-0.0001 (-0.85)	-0.0001 (-1.22)
Degree HHI			-2.4689 (-0.19)	24.5163*** (4.12)	22.2716*** (4.18)	26.7488*** (4.43)	26.1006*** (4.61)	29.8999*** (4.69)
Diameter			0.0026 (0.32)	0.0071 (1.85)	0.0035 (1.09)	0.0061 (1.73)	-0.0023 (-0.71)	-0.0002 (-0.05)
Fragility			0.0061 (0.25)	-0.1496*** (-14.03)	-0.1463*** (-14.01)	-0.1637*** (-13.94)	-0.1441*** (-13.02)	-0.1580*** (-12.72)
Num. Clusters			-0.0052 (-0.31)	0.0069 (0.89)	0.0071 (1.07)	0.0049 (0.68)	0.0051 (0.77)	0.0035 (0.48)
Cluster HHI			-0.3394 (-0.51)	0.2669 (0.84)	0.2240 (0.84)	0.2014 (0.69)	0.1764 (0.67)	0.1566 (0.55)
Control variables included:								
Firm attributes, set 1	No	No	No	No	Yes	No	Yes	No
Firm attributes, set 2	No	No	No	No	No	Yes	No	Yes
U.S. macro variables	No	No	No	No	No	No	Yes	Yes
Observations	4107	4107	4107	4107	3226	2854	3226	2854
R^2	0.470	0.354	0.015	0.820	0.870	0.867	0.873	0.871
Adjusted R^2	0.468	0.352	0.011	0.819	0.869	0.866	0.871	0.869

Panel F: South America								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Constant	2.7790*** (13.59)	2.8187*** (11.76)	3.8847*** (9.20)	2.3978*** (10.05)	-1.7380*** (-10.06)	0.3829* (2.27)	-2.0063*** (-7.63)	0.2076 (0.72)
PD	107.9999*** (13.43)			105.5521*** (18.81)	124.4505*** (24.43)	129.6668*** (24.00)	125.4466*** (24.96)	130.4330*** (24.35)
Degree		0.6814*** (21.31)		0.7838*** (32.89)	0.7037*** (35.97)	0.6879*** (36.21)	0.7061*** (36.63)	0.6901*** (36.74)
Bet. Centrality		0.0001 (0.01)		-0.0039 (-0.96)	-0.0022 (-0.76)	-0.0003 (-0.09)	-0.0024 (-0.84)	-0.0006 (-0.19)
Degree HHI			3.5921 (0.43)	22.9501*** (4.64)	20.6712*** (5.96)	20.4917*** (5.47)	19.3745*** (5.47)	19.4667*** (5.07)
Diameter			-0.0058 (-0.17)	0.0154 (0.80)	0.0075 (0.66)	0.0046 (0.36)	0.0183 (1.49)	0.0124 (0.90)
Fragility			0.0528 (0.69)	-0.4447*** (-9.77)	-0.3955*** (-11.09)	-0.3842*** (-10.31)	-0.3606*** (-9.87)	-0.3491*** (-9.08)
Num. Clusters			-0.0303 (-1.46)	-0.0233 (-1.88)	-0.0215** (-2.82)	-0.0221** (-2.78)	-0.0222** (-2.94)	-0.0229** (-2.86)
Cluster HHI			-0.8368 (-1.64)	-0.6992* (-2.40)	-0.6097*** (-3.73)	-0.6092*** (-3.56)	-0.7515*** (-4.55)	-0.7446*** (-4.30)
Control variables included:								
Firm attributes, set 1	No	No	No	No	Yes	No	Yes	No
Firm attributes, set 2	No	No	No	No	No	Yes	No	Yes
U.S. macro variables	No	No	No	No	No	No	Yes	Yes
Observations	1921	1921	1921	1921	1744	1637	1744	1637
R^2	0.213	0.553	0.103	0.722	0.915	0.908	0.916	0.910
Adjusted R^2	0.208	0.549	0.095	0.719	0.913	0.907	0.915	0.908

Table 7: Summary of adjusted R^2 s from time-series and panel regressions of systemic risk. Time-series regressions are conducted, analogous to the specifications in Table 5, for each regional group; the dependent variables are quarterly changes in regional network level systemic risk score. Panel regressions are conducted, analogous to the specifications in Table 6, for each regional group; the dependent variables are region-specific firms' contribution to network level systemic risk score. Primary explanatory variables include: credit risk (probability of default, PD); and network interconnectedness (degree across all nodes, degree concentration measured by HHI); network parameters (centrality between nodes, diameter, fragility, number of distinct clusters, and HHI concentration within clusters). Additional control variables include firm-specific attributes (book value of assets, market value of equity, loans-to-assets and loans-to-deposits ratios of banks, debt-to-assets and debt-to-equity ratios, debt-to-capital ratio, returns on assets and equity, and market-to-book value of equity) and U.S. macroeconomic variables (default factor, level and slope of term structure factor, TED spread, and VIX). All explanatory variables are in the form of quarterly changes for time-series regressions and in the form of quarterly level values for panel regressions.

Included explanatory variables	Regional Group					
	All	East Asia	South Asia	Eastern Europe	South Europe & Africa	South America
Panel A: Adjusted R^2 s from time-series regressions						
Credit risk (only)		48%	40%	76%	32%	0%
Network parameters (only)						
Firm-level		87%	57%	59%	76%	93%
Network-level		86%	57%	57%	73%	90%
Credit risk + network parameters		94%	81%	90%	95%	97%
Credit risk + network parameters + firm-specific attributes		96-97%	91-94%	93-95%	96%	99%
Credit risk + network parameters + firm-specific attributes + U.S. macro variables		98%	97%	96-97%	96%	99%
Panel B: Adjusted R^2 s from panel regressions						
Credit risk (only)	9%	21%	50%	48%	47%	21%
Network parameters (only)						
Firm-level	4%	61%	38%	64%	35%	55%
Network-level	47%	8%	4%	41%	1%	10%
Credit risk + network parameters	64%	75%	80%	72%	82%	72%
Credit risk + network parameters + firm-specific attributes	68%	82-83%	85%	86-87%	87%	91%
Credit risk + network parameters + firm-specific attributes + U.S. macro variables	68%	82-83%	85%	87%	87%	91-92%

Table 8: Correlations of quarterly changes in systemic risk between country regional groups. Systemic risk is the regional network level systemic risk score. Panel A reports contemporaneous correlations of changes in systemic risk, Panel B reports one-quarter lagged correlations of changes in systemic risk. Lagged values are on the columns and contemporaneous ones are on the rows. p -values are reported in parentheses.

	East Asia	South Asia	Eastern Europe	South Europe & Africa	South America
Panel A: Contemporaneous correlations					
East Asia	1.0000				
South Asia	0.1744 (0.2308)	1.0000			
Eastern Europe	0.4970 (0.0003)	−0.0001 (0.9997)	1.0000		
South Europe & Africa	0.3394 (0.0170)	−0.1972 (0.1744)	0.2438 (0.0915)	1.0000	
South America	0.4422 (0.0015)	0.0553 (0.7057)	0.5573 (0.0000)	0.2507 (0.0823)	1.0000
Panel B: Lagged correlations					
East Asia	0.1196 (0.4180)	−0.1737 (0.2378)	−0.0717 (0.6284)	0.0652 (0.6597)	−0.1732 (0.2392)
South Asia	0.0504 (0.7336)	−0.0266 (0.8574)	0.1442 (0.3280)	0.4079 (0.0040)	−0.0350 (0.8135)
Eastern Europe	0.6675 (0.0000)	−0.0144 (0.9226)	0.2901 (0.0455)	0.2876 (0.0475)	0.2407 (0.0993)
South Europe & Africa	0.1621 (0.2711)	−0.0631 (0.6703)	−0.0646 (0.6624)	−0.1276 (0.3873)	0.0293 (0.8435)
South America	0.3813 (0.0075)	−0.0527 (0.7219)	−0.2145 (0.1432)	−0.0102 (0.9452)	−0.3351 (0.0199)

Table 9: Granger causality regressions. For each country regional group, quarterly changes in systemic risk measure (regional network level systemic risk score) are regressed on one-quarter lagged changes in systemic risk measures of all five regional groups (including itself). Panel A reports the univariate F -statistics of significance (and corresponding p -values in parentheses). Panel B reports the joint F -statistics of significance (and corresponding p -values in parentheses) for the four other cross regional groups considered together.

Explanatory variables: lagged systemic risk of	Dependent variable: systemic risk corresponding to				
	East Asia	South Asia	Eastern Europe	South Europe & Africa	South America
Panel A: Univariate F -statistics of lagged variables					
East Asia	2.92 (0.0948)	0.69 (0.4111)	26.32 (0.0000)	3.81 (0.0578)	42.39 (0.0000)
South Asia	1.95 (0.1695)	0.52 (0.4730)	1.21 (0.2767)	1.33 (0.2545)	3.41 (0.0717)
Eastern Europe	0.22 (0.6436)	1.61 (0.2111)	0.12 (0.7339)	1.36 (0.2498)	6.44 (0.0150)
South Europe & Africa	0.00 (0.9908)	9.98 (0.0029)	0.11 (0.7374)	2.33 (0.1346)	1.36 (0.2495)
South America	1.96 (0.1694)	1.71 (0.1975)	0.16 (0.6893)	0.13 (0.7229)	14.04 (0.0005)
Panel B: Joint F -statistic of all four lagged cross-variables					
All 4 lagged cross-variables	1.32 (0.2790)	3.04 (0.0276)	7.67 (0.0001)	1.19 (0.3293)	10.79 (0.0000)

Table 10: Vector autoregression, VAR (significant results only). Quarterly changes in systemic risk measure (regional network level systemic risk score) of the five country regional groups are jointly regressed on four lagged (one through four quarters) changes in systemic risk measure of the five regional groups. Both likelihood ratio (LR) and Akaike information criterion (AIC) identify that a maximum of three lags are material. Out of 100 explanatory variables (5 regressions * 5 systemic risk scores * 4 lags), only 8 are significant at 5% level. The following summary reports the coefficients and t -statistics (corresponding p -values in parentheses) for these significant 8.

Dependent variable: systemic risk of	Explanatory variable: systemic risk of	Lag in quarters	Coefficient	t -statistic (p -value)
East Asia	South Europe & Africa	2	0.5509	2.51 (0.019)
South Asia	South Europe & Africa	1	0.4160	2.12 (0.045)
South Asia	South Europe & Africa	4	-0.8036	-3.03 (0.006)
South Asia	South America	3	-0.3840	-2.18 (0.039)
Eastern Europe	East Asia	1	1.3856	2.96 (0.007)
South America	East Asia	1	1.6759	3.93 (0.001)
South America	South America	1	-0.7560	-3.33 (0.003)
South America	South America	2	-0.5788	-2.12 (0.045)

Table 11: Principal component analysis. Panel A (Panel B) presents the first five principal components and corresponding eigen values for quarterly levels of (changes in) the systemic risk measure (network level systemic risk score) of the five regional country groups. Panel C reports the results of time-series regressions of the quarterly changes in first three principal components corresponding to quarterly levels of systemic risk on quarterly changes in U.S. macroeconomic variables (default factor, level and slope of term structure factor, TED spread, and VIX); all regressions include adjustments for heteroscedasticity, autocorrelation in residuals and year fixed-effects. t -statistics are reported in parentheses. Coefficients marked by *, **, and *** denote $p < 0.05$, $p < 0.01$, and $p < 0.001$, respectively.

Panel A: First 5 principal components of levels of systemic risk				
Component	Eigenvalue	Difference	Proportion	Cumulative
1	2.6124	1.6143	0.5225	0.5225
2	0.9981	0.0245	0.1996	0.7221
3	0.9737	0.7293	0.1947	0.9169
4	0.2443	0.0730	0.0489	0.9657
5	0.1714		0.0343	1.0000

Panel B: First 5 principal components of changes in systemic risk				
Component	Eigenvalue	Difference	Proportion	Cumulative
1	2.1944	1.0355	0.4389	0.4389
2	1.1589	0.4112	0.2318	0.6707
3	0.7477	0.2512	0.1495	0.8202
4	0.4965	0.0941	0.0993	0.9195
5	0.4025		0.0805	1.0000

Panel C: Regression of changes in first 3 principal components of levels of systemic risk			
Explanatory variables	Dependent variable		
	Component 1	Component 2	Component 3
Constant	-1.2608*** (-3.65)	0.0882 (0.25)	-1.8312*** (-8.03)
Default	2.3225* (2.31)	-0.1131 (-0.27)	0.9731 (1.68)
Term (Level)	0.0449 (0.08)	0.3639 (1.16)	-0.1814 (-0.49)
Term (Slope)	-0.3728 (-0.59)	-0.1903 (-0.34)	0.4955 (1.21)
TED	0.6894 (1.52)	-0.2168 (-0.90)	0.8615*** (4.29)
VIX	-0.0597 (-1.43)	0.0064 (0.17)	-0.0449 (-1.61)
Observations	49	49	49
R^2	0.519	0.354	0.458
Adjusted R^2	0.255	-0.000	0.160

Table 12: Predictive time-series regressions of quarterly changes in credit risk. Panels A through E report the results for individual geographical regions. The dependent variable is the difference in values of credit risk (mean probability of default, PD) between quarters t and $t - 1$. Primary explanatory variables include: value of probability of default in quarter $t - 1$; quarterly changes in systemic risk (network level systemic risk score); and quarterly changes in network parameters (mean degree across all nodes, mean centrality between nodes, degree concentration measured by HHI, diameter, fragility, number of distinct clusters, and HHI concentration within clusters). Additional control variables include: quarterly changes in median firm-specific attributes, set 1 (book value of assets, loans-to-assets and loans-to-deposits ratios of banks, debt-to-assets ratios, debt-to-capital ratio, returns on assets, and market-to-book value of equity); quarterly changes in median firm-specific attributes, set 2 (market value of equity, loans-to-assets and loans-to-deposits ratios of banks, debt-to-equity ratios, debt-to-capital ratio, returns on equity, and market-to-book value of equity); and quarterly changes in U.S. macroeconomic variables (default factor, level and slope of term structure factor, TED spread, and VIX). Mean PD, degree and centrality across firms, and market-wide median firm-specific attributes are computed every quarter. All regressions include adjustments for heteroskedasticity, autocorrelation in residuals and year fixed-effects. t -statistics are reported in parentheses. Coefficients marked by *, **, and *** denote $p < 0.05$, $p < 0.01$, and $p < 0.001$, respectively.

Panel A: East Asia									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Constant	-0.0003 (-0.42)	-0.0010*** (-4.71)	-0.0008 (-1.23)	-0.0005 (-1.07)	-0.0005 (-0.86)	-0.0009 (-1.07)	-0.0014 (-1.68)	-0.0012** (-3.15)	-0.0015*** (-4.14)
Mean PD	-0.1070 (-1.22)		-0.0258 (-0.32)	-0.0048 (-0.10)	0.0005 (0.01)	0.0636 (0.76)	0.1421 (1.62)	0.0926 (1.78)	0.1374** (3.19)
Systemic Risk		0.0008* (2.50)	0.0008 (1.92)	0.0026*** (4.65)	0.0026*** (4.69)	0.0027** (3.22)	0.0029*** (4.03)	0.0020** (3.31)	0.0018* (2.91)
Mean Degree				-0.0004** (-3.45)	-0.0004 (-1.10)	-0.0007* (-2.46)	-0.0006* (-2.66)	-0.0003 (-1.14)	-0.0003 (-1.57)
Mean Bet. Centrality				0.0000 (0.31)	0.0000 (0.39)	0.0000 (0.34)	-0.0000 (-0.01)	0.0000 (1.67)	0.0000 (0.25)
Degree HHI					0.0612 (0.27)	-0.0189 (-0.06)	0.0206 (0.08)	0.3173 (1.23)	0.2032 (0.87)
Diameter					0.0000 (0.60)	-0.0000 (-0.03)	0.0000 (0.53)	-0.0000 (-1.07)	-0.0000 (-0.01)
Fragility					-0.0000 (-0.16)	0.0001 (0.40)	0.0000 (0.32)	-0.0001 (-0.36)	-0.0000 (-0.12)
Num. Clusters					0.0000 (0.15)	-0.0001 (-0.72)	-0.0001 (-0.61)	-0.0003*** (-4.41)	-0.0003** (-3.65)
Cluster HHI					0.0005 (0.05)	-0.0068 (-0.53)	-0.0065 (-0.58)	-0.0247** (-3.30)	-0.0204* (-2.76)
Control variables included:									
Firm attributes, set 1	No	No	No	No	No	Yes	No	Yes	No
Firm attributes, set 2	No	No	No	No	No	No	Yes	No	Yes
U.S. macro variables	No	No	No	No	No	No	No	Yes	Yes
Observations	49	49	49	49	49	49	49	49	49
R^2	0.786	0.844	0.844	0.914	0.913	0.953	0.959	0.996	0.996
Adjusted R^2	0.706	0.786	0.780	0.871	0.845	0.888	0.900	0.987	0.988

Panel B: South Asia									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Constant	0.0030 (1.75)	-0.0001 (-0.09)	0.0041** (3.53)	0.0032* (2.58)	0.0012 (0.81)	0.0027 (2.00)	0.0045** (2.87)	0.0006 (0.43)	0.0041** (3.48)
Mean PD	-0.5849* (-2.51)		-0.5551*** (-3.81)	-0.2773 (-1.54)	-0.2288 (-1.55)	-0.3037* (-2.15)	-0.6447*** (-5.85)	-0.2265 (-1.47)	-0.6686*** (-5.38)
Systemic Risk		0.0018 (1.52)	0.0023** (3.15)	0.0047*** (5.79)	0.0044*** (6.04)	0.0014 (1.36)	0.0044*** (5.32)	0.0016 (0.84)	0.0059*** (6.67)
Mean Degree				-0.0007** (-3.53)	-0.0018* (-2.22)	-0.0013 (-1.73)	-0.0003 (-0.31)	-0.0020* (-2.58)	-0.0004 (-0.72)
Mean Bet. Centrality				0.0000 (0.39)	0.0000 (0.63)	-0.0000 (-0.93)	0.0000 (0.99)	-0.0000 (-0.99)	0.0000* (2.29)
Degree HHI					-0.4774 (-1.09)	0.3117 (0.84)	-0.3045 (-0.45)	-0.3194 (-0.67)	-0.6694 (-1.34)
Diameter					0.0001 (0.85)	0.0003* (2.73)	0.0002 (1.71)	0.0002* (2.73)	0.0002 (1.98)
Fragility					0.0007 (1.53)	0.0005 (1.22)	-0.0000 (-0.09)	0.0010 (2.11)	-0.0001 (-0.33)
Num. Clusters					0.0012* (2.69)	0.0008* (2.21)	0.0012* (2.81)	0.0010* (2.41)	0.0009* (2.24)
Cluster HHI					0.0820* (2.50)	0.0624* (2.39)	0.1034** (2.88)	0.0692 (2.12)	0.0842* (2.40)
Control variables included:									
Firm attributes, set 1	No	No	No	No	No	Yes	No	Yes	No
Firm attributes, set 2	No	No	No	No	No	No	Yes	No	Yes
U.S. macro variables	No	No	No	No	No	No	No	Yes	Yes
Observations	49	49	49	49	49	49	49	49	49
R^2	0.601	0.292	0.756	0.852	0.912	0.955	0.974	0.980	0.991
Adjusted R^2	0.452	0.028	0.655	0.778	0.844	0.892	0.938	0.937	0.971

Panel C: Eastern Europe									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Constant	-0.0003 (-0.30)	0.0021** (3.19)	0.0021* (2.33)	0.0021*** (4.53)	0.0018 (0.59)	0.0006 (0.15)	0.0013 (0.37)	-0.0065 (-1.38)	-0.0075 (-2.04)
Mean PD	0.0367 (0.11)		0.0109 (0.06)	0.0220 (0.27)	0.0300 (0.32)	0.0381 (0.37)	0.0712 (0.78)	0.1259 (1.69)	0.1645* (2.43)
Systemic Risk		0.0034*** (3.65)	0.0034*** (3.65)	0.0050*** (6.19)	0.0048*** (6.20)	0.0053*** (5.17)	0.0056*** (5.04)	0.0020 (1.01)	0.0019 (1.42)
Mean Degree				-0.0023*** (-3.63)	-0.0012 (-0.83)	-0.0013 (-0.43)	-0.0018 (-0.56)	0.0005 (0.13)	0.0020 (0.80)
Mean Bet. Centrality				0.0000 (0.79)	-0.0000 (-0.44)	-0.0000 (-0.57)	-0.0000 (-0.36)	0.0001 (0.89)	0.0000 (0.66)
Degree HHI					-0.0033 (-0.09)	0.0058 (0.09)	-0.0086 (-0.19)	0.1081 (1.57)	0.1173 (2.10)
Diameter					0.0001 (0.36)	0.0001 (0.38)	-0.0000 (-0.05)	-0.0001 (-0.27)	-0.0000 (-0.20)
Fragility					0.0001 (0.16)	-0.0001 (-0.09)	0.0004 (0.46)	-0.0010 (-0.61)	-0.0012 (-1.05)
Num. Clusters					0.0001 (0.97)	0.0001 (0.72)	0.0001 (0.82)	-0.0002 (-0.99)	-0.0002 (-1.32)
Cluster HHI					-0.0009 (-0.23)	0.0003 (0.06)	-0.0002 (-0.03)	-0.0064 (-0.90)	-0.0086 (-1.58)
Control variables included:									
Firm attributes, set 1	No	No	No	No	No	Yes	No	Yes	No
Firm attributes, set 2	No	No	No	No	No	No	Yes	No	Yes
U.S. macro variables	No	No	No	No	No	No	No	Yes	Yes
Observations	49	49	49	49	47	47	47	47	47
R^2	0.565	0.827	0.828	0.914	0.926	0.953	0.962	0.985	0.987
Adjusted R^2	0.404	0.763	0.757	0.871	0.865	0.880	0.903	0.946	0.955

Panel D: South Europe & Africa									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Constant	0.0013 (1.40)	-0.0003 (-0.66)	0.0009 (1.09)	0.0011* (2.59)	0.0011* (2.76)	0.0020** (3.12)	0.0020** (3.62)	0.0027*** (6.50)	0.0026*** (6.50)
Mean PD	-0.3392 (-1.95)		-0.2387 (-1.90)	-0.1930*** (-4.17)	-0.2058*** (-3.97)	-0.2328** (-2.89)	-0.2154** (-2.91)	-0.3566*** (-6.86)	-0.3929*** (-7.76)
Systemic Risk		0.0015*** (3.67)	0.0012** (2.85)	0.0056*** (11.49)	0.0060*** (11.94)	0.0062*** (11.84)	0.0058*** (8.48)	0.0047*** (9.88)	0.0040*** (4.69)
Mean Degree				-0.0021*** (-11.00)	-0.0026*** (-4.09)	-0.0017 (-1.89)	-0.0015 (-1.81)	0.0002 (0.22)	0.0005 (0.75)
Mean Bet. Centrality				0.0000 (1.09)	-0.0000 (-0.03)	0.0000 (0.30)	0.0000 (0.38)	-0.0000 (-1.80)	-0.0000 (-1.37)
Degree HHI					-0.0746 (-1.07)	-0.0494 (-0.60)	-0.0175 (-0.23)	0.0890 (1.45)	0.1173 (2.02)
Diameter					-0.0000 (-0.21)	-0.0000 (-0.96)	-0.0000 (-0.69)	0.0001 (1.15)	0.0001 (1.15)
Fragility					0.0002 (0.49)	-0.0003 (-0.59)	-0.0003 (-0.71)	-0.0008 (-2.08)	-0.0009* (-2.39)
Num. Clusters					0.0000 (0.27)	0.0001 (0.95)	0.0001 (0.82)	0.0001 (1.64)	0.0002* (2.77)
Cluster HHI					0.0015 (0.51)	0.0030 (0.90)	0.0027 (0.90)	0.0046 (1.51)	0.0067* (2.64)
Control variables included:									
Firm attributes, set 1	No	No	No	No	No	Yes	No	Yes	No
Firm attributes, set 2	No	No	No	No	No	No	Yes	No	Yes
U.S. macro variables	No	No	No	No	No	No	No	Yes	Yes
Observations	49	49	49	49	49	49	49	49	49
R^2	0.482	0.562	0.591	0.935	0.956	0.970	0.973	0.992	0.992
Adjusted R^2	0.290	0.399	0.422	0.902	0.922	0.929	0.935	0.974	0.973

Panel E: South America									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Constant	-0.0012 (-1.10)	-0.0008*** (-4.64)	-0.0013 (-1.06)	-0.0020* (-2.42)	-0.0014* (-2.43)	-0.0018* (-2.46)	-0.0011 (-1.60)	-0.0020* (-2.66)	-0.0018** (-3.38)
Mean PD	0.0551 (0.41)		0.0625 (0.42)	0.1900 (1.90)	0.1187 (1.65)	0.1365 (1.64)	0.0574 (0.72)	0.1989 (2.12)	0.1764* (2.64)
Systemic Risk		-0.0002 (-0.96)	-0.0002 (-0.91)	0.0017*** (3.67)	0.0022*** (6.56)	0.0024*** (8.49)	0.0025*** (8.75)	0.0020*** (6.70)	0.0021*** (7.93)
Mean Degree				-0.0014** (-3.59)	-0.0002 (-0.27)	-0.0001 (-0.21)	-0.0003 (-0.44)	-0.0002 (-0.26)	-0.0004 (-0.69)
Mean Bet. Centrality				0.0000 (0.51)	0.0001* (2.19)	0.0000 (1.27)	0.0000 (0.94)	0.0000 (1.64)	0.0000 (1.56)
Degree HHI					0.0400** (3.28)	0.0501* (2.44)	0.0431* (2.09)	0.0432 (1.75)	0.0362 (1.85)
Diameter					-0.0001** (-2.83)	-0.0001* (-2.52)	-0.0001 (-1.70)	-0.0001** (-3.05)	-0.0001* (-2.94)
Fragility					-0.0010*** (-3.94)	-0.0011* (-2.75)	-0.0010* (-2.59)	-0.0009 (-1.86)	-0.0008 (-1.98)
Num. Clusters					0.0001* (2.16)	0.0001 (2.09)	0.0001 (1.56)	0.0000 (1.31)	0.0000 (1.27)
Cluster HHI					0.0016** (3.08)	0.0020** (3.17)	0.0016* (2.22)	0.0011** (2.96)	0.0012* (2.56)
Control variables included:									
Firm attributes, set 1	No	No	No	No	No	Yes	No	Yes	No
Firm attributes, set 2	No	No	No	No	No	No	Yes	No	Yes
U.S. macro variables	No	No	No	No	No	No	No	Yes	Yes
Observations	49	49	49	49	49	49	49	49	49
R^2	0.447	0.423	0.459	0.811	0.947	0.978	0.976	0.991	0.991
Adjusted R^2	0.242	0.209	0.236	0.717	0.907	0.947	0.942	0.972	0.972

Table 13: Summary of adjusted R^2 s from predictive time-series regressions of quarterly changes in credit risk. Time-series regressions are conducted, analogous to the specifications in Table 12, for each regional group; the dependent variables are differences in values of credit risk (mean probability of default, PD) between quarters t and $t - 1$. Primary explanatory variables include: value of probability of default in quarter $t - 1$; quarterly changes in systemic risk (network level systemic risk score); and quarterly changes in network parameters (mean degree across all nodes, mean centrality between nodes, degree concentration measured by HHI, diameter, fragility, number of distinct clusters, and HHI concentration within clusters). Additional control variables include firm-specific attributes (book value of assets, market value of equity, loans-to-assets and loans-to-deposits ratios of banks, debt-to-assets and debt-to-equity ratios, debt-to-capital ratio, returns on assets and equity, and market-to-book value of equity) and U.S. macroeconomic variables (default factor, level and slope of term structure factor, TED spread, and VIX).

Included explanatory variables	Regional Group				
	East Asia	South Asia	Eastern Europe	South Europe & Africa	South America
Lagged credit risk (only)	71%	45%	40%	29%	24%
Systemic risk (only)	79%	3%	76%	40 %	21%
Lagged credit risk + systemic risk	78%	66%	76%	42%	24%
Lagged credit risk + systemic risk + network parameters	85-87%	78-84%	87%	90-92%	72-91%
Lagged credit risk + systemic risk + network parameters + firm-specific attributes	89-90%	89-94%	88-90%	93-94%	94-95%
Lagged credit risk + systemic risk + network parameters + firm-specific attributes + U.S. macro variables	99%	94-97%	95-96%	97%	97%